Spark AR

Spark AR Certification Study Guide



OVERVIEW

The goal of this study aid is to educate, motivate and nurture learners as they prepare to take the Spark AR Certification exam. To make the most of your experience, we recommend that you use this study aid in conjunction with the latest version of Spark AR Studio. To download the latest version of Spark AR, please click here.

Click here to learn more about Facebook Blueprint Certifications.

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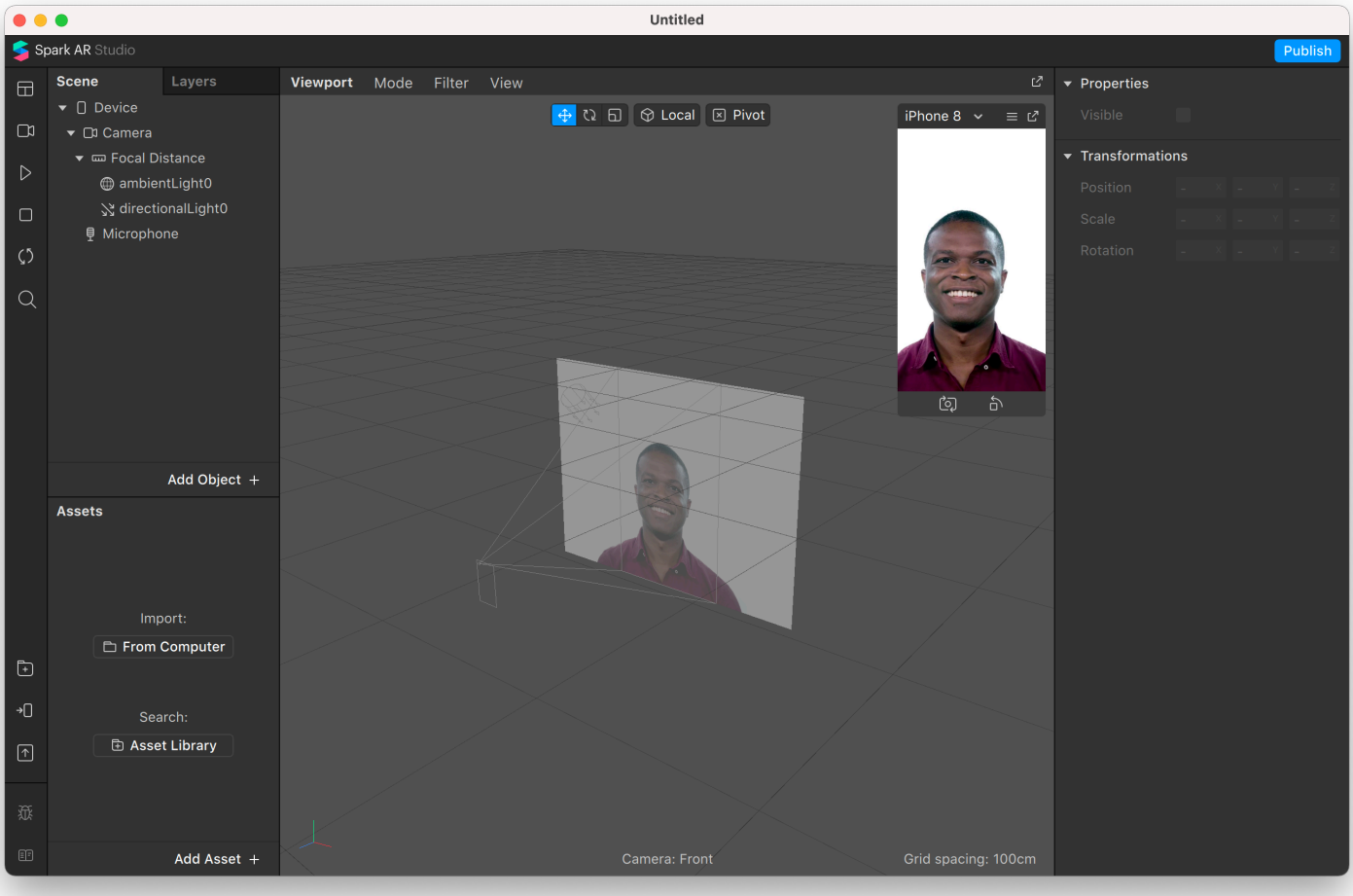
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Glossary

AN INTRODUCTION TO SPARK AR



What can you do on the Spark AR platform?

Build with or without code.

You don't have to be an expert coder to create effects in Spark AR. Spark AR lets you use visual programming to easily add interactivity, animate objects and build logic into your effects. You can even make textures and materials to use in your scene.

Import objects and sounds.

Add your own sound files and 3D objects, browse Facebook's free library of assets or import high-fidelity models through our partnership with Sketchfab.

Publish effects to Instagram.

Spark AR Hub helps you publish, manage and understand your AR effect performance across Facebook and Instagram.

Spark AR Studio provides tools you can use to create stunning augmented reality experiences.

With Spark AR you can:

- Create materials, textures and some 3D and 2D objects in Spark AR Studio. You'll also need to import your own.
- Create effects that respond to someone's face and body.
- Create effects that transform someone's world.
- Import your own 3D objects, including ones with bones and joints, and animations.
- Create and position 2D surfaces, and apply textures to them.
- Add and edit 2D and 3D text, including dynamic text.
- Add depth to objects in your scene with 5 different types of lighting.
- Add audio clips, or create distortion effects using the microphones in mobile devices.

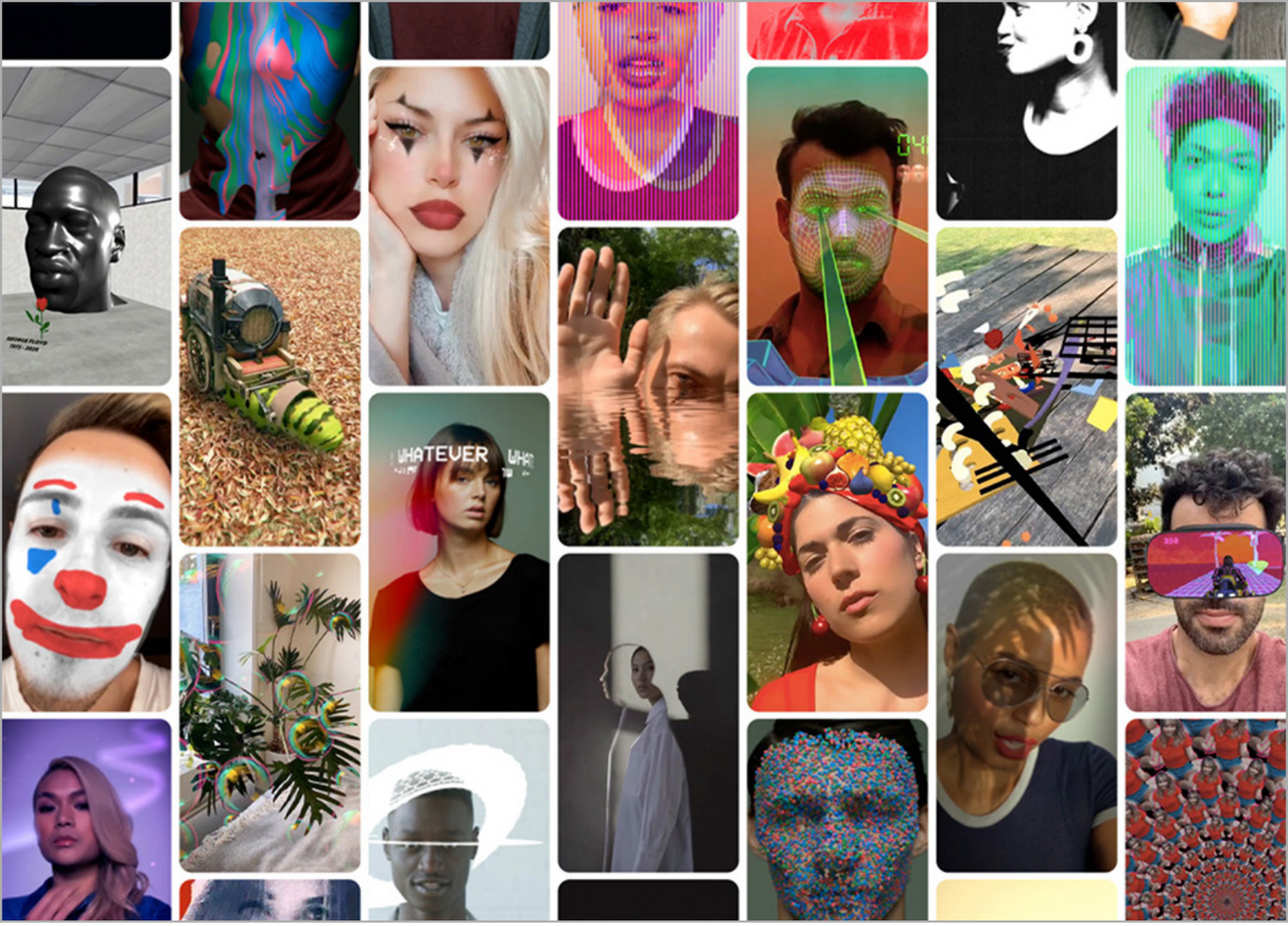
- Use particle systems to generate, display and move particles in great numbers.
- Use multiple texture inputs to apply a realistic look to the surface of your objects.
- Use the Patch Editor to create interactive experiences without scripting. Or, create effects using JavaScript.
- Use the LiveStreaming module to get information if somebody uses your effect when they go live on Facebook.
- Use the Spark AR Player app to preview and test your effect.

This study aid can help you improve your skills as a Spark AR content creator as you prepare for the Spark AR Certification exam.

Start your journey in augmented reality with **Chapter 1: Design & Concepting**.

Chapter 1

DESIGN & CONCEPTING



Get Started with Spark AR

Whether you're a new learner or an experienced professional, Spark AR Studio has all of the features and capabilities you need in order to create engaging effects for clients.

In this chapter you'll learn about the process of making an effect in Spark AR Studio and the features you can use to create stunning augmented reality experiences.



YOU CAN USE SPARK AR TO CREATE ENGAGING AUGMENTED REALITY EFFECTS, BUT BEFORE YOU START DEVELOPING AN EFFECT, IT'S HELPFUL TO BECOME FAMILIAR WITH A FEW BEST PRACTICES.

Focus on a solid use case.
You should focus on a specific goal or work toward addressing a specific task. For example, do you want to create an effect that expresses a feeling? Or perhaps you want to address current news, topics or trends.

Make it social.
Consider adding support for multiple faces to your effect, so people can use it with friends.

Design for different device types.
Test your effect on multiple types and generations of mobile devices to make sure that effect works well.

Design for repeat use.
The most engaging effects are ones that are flexible and can be used in different contexts. Effects that are relevant year-round will give your work a longer shelf life.

Keep it simple.
Not all people will have experienced AR before. To stop people becoming frustrated, manage expectations on what they can expect from the experience and how much effort they'll need to contribute. You can do this by:

- Adding clear instructions.
- Making sure any interactions are intuitive.
- Focusing on one or two main interactions, so people can learn the full effect experience easily.

Take note:
When planning your effect, keep in mind that effects submitted to both Facebook and Instagram are reviewed to make sure they follow our community standards and platform policies.

When designing your effect you'll want to:
Add to the camera experience.
Avoid taking over lots of the camera view, or making people completely unrecognizable. Allowing people to keep recognizable elements of their environment or body helps make the effect feel personal.

Avoid the edges of the screen.
If you add objects on the edges of the composition, there's a chance they will be obscured by UI. It's best to avoid placing objects too close to the edge of the camera screen if you can.

Give people something to capture with the lowest amount of effort.
Most people move on from an effect in a few seconds. Make sure the main features of your effect happen as soon as possible.

Keep interactions obvious.
Think of the most natural movement to trigger the most obvious change – for example, pinching the screen to make an object bigger or smaller.

Avoid using gestures that conflict with system-level commands.
These are interactions that people are already familiar with on their mobile devices. Changing how these commands work can be confusing and frustrating. System-level commands include:

- **Switch camera view** - double tap.
- **Camera zoom** - pinch in/out (except when resizing stickers or 3D objects).
- **Video zoom** - touch and hold camera capture button then drag up.
- **Return to Facebook News Feed** - swipe left.
- **Keep interactions to a minimum.** Focus on 1 or 2 main interactions, so that people can learn the full effect experience easily.

Pro Tip: As you're scoping for your client's effect, be mindful of accessibility concerns for the end user. For example, if your effect is going to contain flashing lights you might consider adding a warning so that it's appropriate for all audiences.

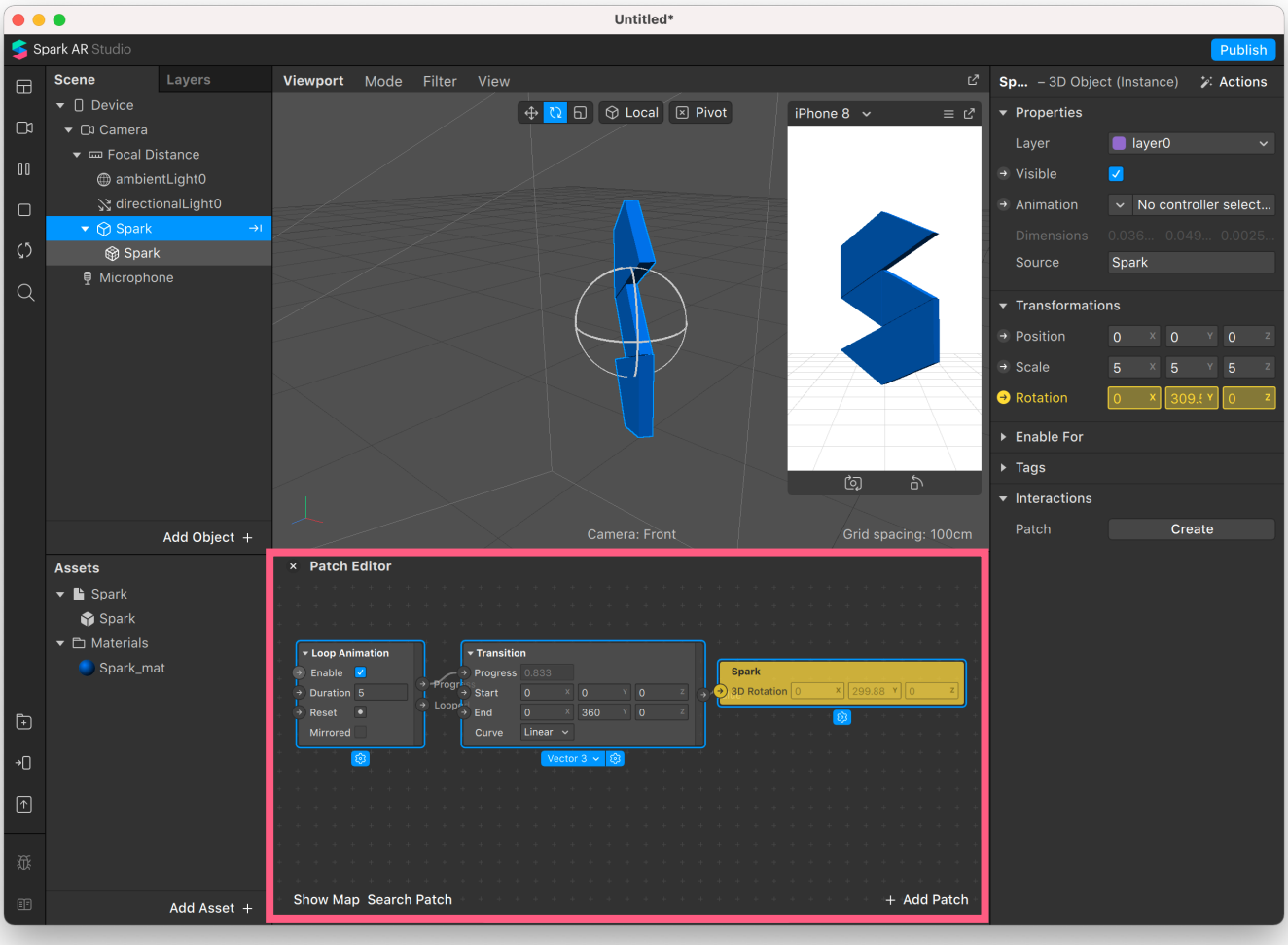
TO CREATE EFFECTS IN SPARK AR STUDIO, YOU WILL WORK WITH OBJECTS AND ASSETS.

There are a range of objects available in Spark AR Studio which you can use to add all kinds of elements to your effect. Some of these include 2D and 3D objects, sounds, typography and lights.

You can either make assets in Spark AR Studio or you can import them from your computer and add them to your project through the Assets Panel. Keep in mind that you can import one or more at a time. Once imported, your assets are listed in the Assets Panel. When you select an asset, you're able to make changes to it in the Inspector panel.

Spark AR Studio supports various file formats, depending on the type of object or asset you're importing:

2D Assets	3D Models	Audio	Fonts
PNG JPEG SVG	FBX 2014/2015 (binary and ASCII versions) gITF 2 (binary and text versions) COLLADA / DAE OBJ DAE The following features are supported for 3D models: • 3D scene • Materials • Textures • Animations targeting a model's position, rotation and scale	Mono M4A AAC codec, with a sampling frequency of 44.1KHz	TrueType/ OpenType



Patch editor

THE PATCH EDITOR

Patches in Spark AR Studio let you add interaction, animation and logic to effects. You can also use them to create materials. Patches act as visual building blocks, with each patch performing a different function. Connected together in what's known as a graph, patches pass and receive information to and from one another to make things happen in the effect. Many capabilities available in scripting can be added to your effect as patches, so you can add complexity without knowing how to code.

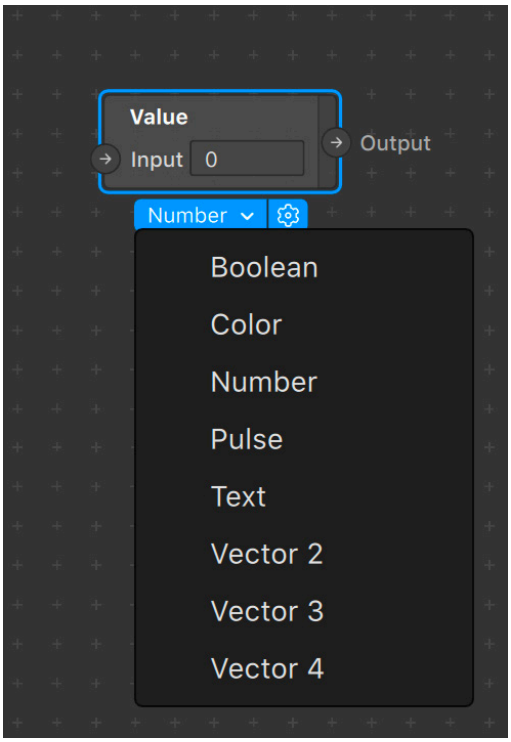
Most scene objects and assets can be represented as patches. You can also create patches to represent an object or an asset's properties in the Inspector. These patches can be used to add all kinds of additional complexity to your project, including logic, animation and interactivity.

Patch Editor data types:

- **Boolean Signal** – at any point in time, a boolean signal is either true or false. For example, the Mouth Open port in the Mouth Open patch can tell whether a mouth is either open, or closed.
- **Scalar** – a scalar is a one-dimensional value. The Mouth Openness port in the Mouth Open patch has a scalar value because it reflects how open the mouth is. If a mouth is wide open the value might be 1, if it's half open it might be 0.5.
- **Pulse** – a pulse is an event that happens at some point in time. It's not a continuous signal like a boolean – which is always either true or false.
- **3D Point** – a 3D point is a three-dimensional signal that can represent a rotation, scale or position. For example, a patch representing the position of a 3D object has a 3D Position port. This port has a 3D

Point data type, because the object is positioned in 3D space.

- **2D Point** – A 2D point is a two-dimensional signal that can represent a rotation, scale or position. Patches with a 2D point type have X and Y values.
- **Color** – this type represents the RGBA channels – Red, Green, Blue and Alpha. It's indicated by a square with a color.
- **Progress** – progress is a scalar signal that's restricted to the 0–1 range. It could be used to drive an animation. Ports with progress types are labeled Progress, like in the Animation patch.
- **Vector** – VectorN is a multidimensional value consisting of N scalar components.
  - Vector2 for 2D points and 2D texture coordinates.
  - Vector3 for 3D points, 3D normals and 3-channel (RGB) color.
  - Vector4 for 4D position and 4-channel (RGBA) color.
- **Matrix** – MatrixN represents a matrix of N by N scalar components.
  - Matrix2 to represent a rotation transformation for 2D vectors.
  - Matrix3 to represent a rotation transformation for 3D vectors or general affine transformation for 2D vectors with an additional homogeneous coordinate, which effectively makes them Vector3.
  - Matrix4 to represent a general affine transformation for 3D vectors with an additional homogeneous coordinate, which effectively makes them Vector4.
- **Texture 2D** – Texture2D typically represents an image and can have 1 to 4 channels.



Value patch data types

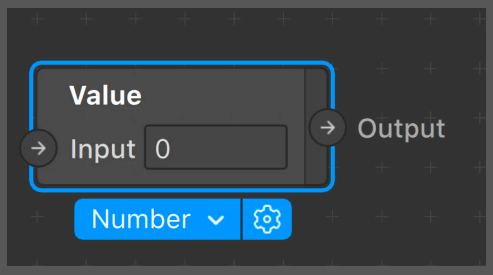
When a port has a box next to it, it means you can edit the value associated with the port. Some patches are able to switch their data types to accommodate a creator's needs, as in the case of the 'Value Patch' that can be used as a color or a number.

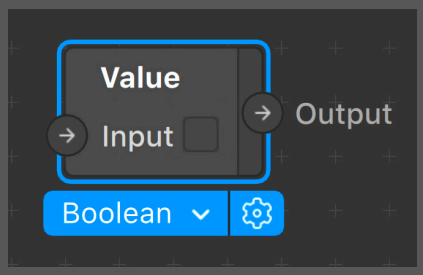
You can choose different data types for some of the patches. For example, if you're manipulating the shape of a 2D object, you'd set the data type of certain patches to Vector 2. If you're applying a color to a material, you'd change the data type of certain patches to Color.

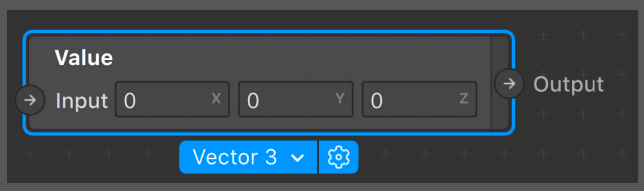
Pro-tip

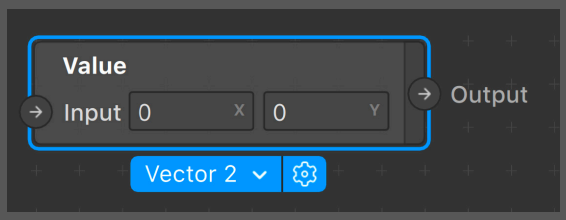
Note how Spark AR UI changes according to the data type. In this illustration you can see the different UI using the value patch. This process can be helpful when you create the patch graph and need to figure out what patches come next and can also help you understand how to debug with patches.

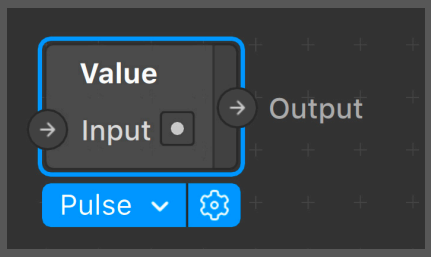
In the example below, the UI of the number data type default value is 0 and it expects a number input. The UI of the boolean data type is a checkbox, which should be ticked if the input is true and unticked if the input is false.

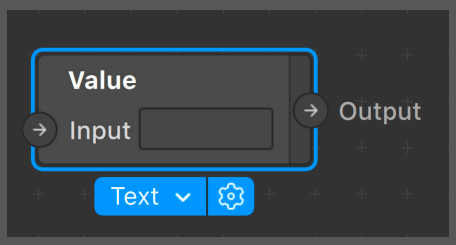


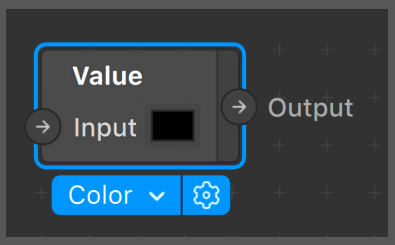




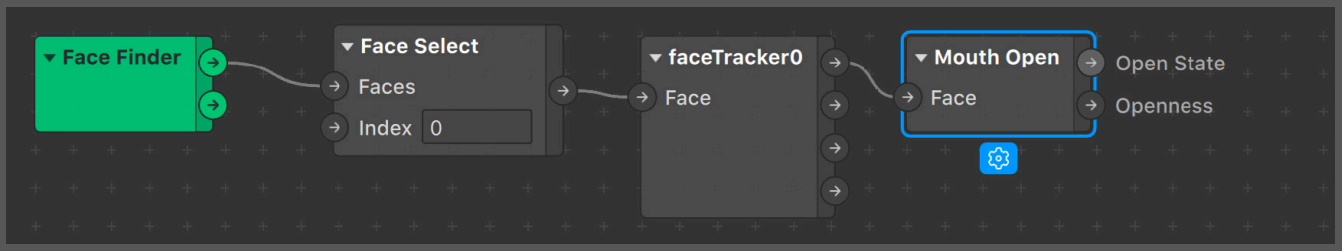




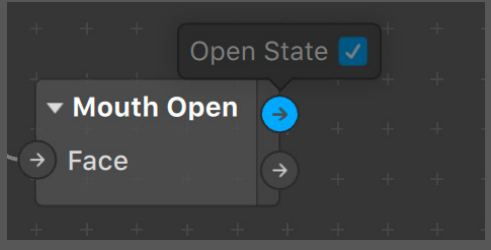


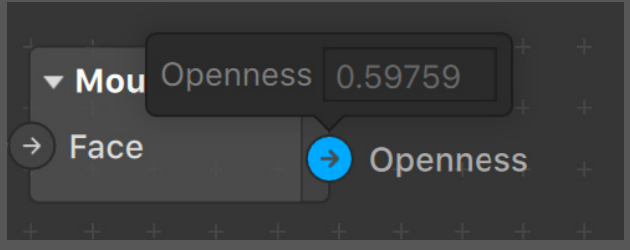


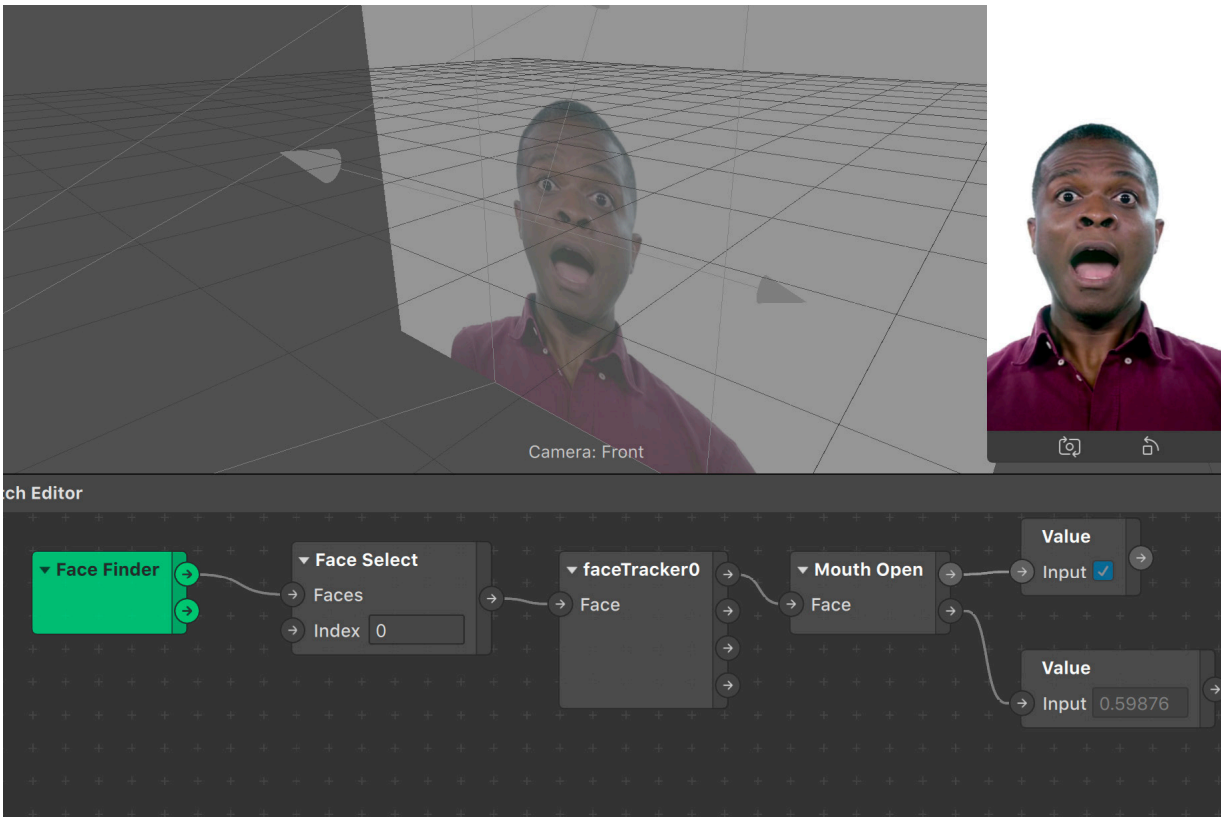
In this example, the mouth open patch has two outputs, 'open state' which is boolean and is either true or false, and 'openness', which is a number from 0 to 1 that indicates how open the mouth is.



An intuitive way to create the patch graph is to look at the UI data type of the output or to use a value patch to debug and to understand the connection flow .







When the face tracker detects mouth open, open state is checked as true and the openness value range from 0 to 1

What are patches?

Patches allow you to add interaction and animation to your effects. Each patch performs a unique function and can pass and receive information to and from other patches.

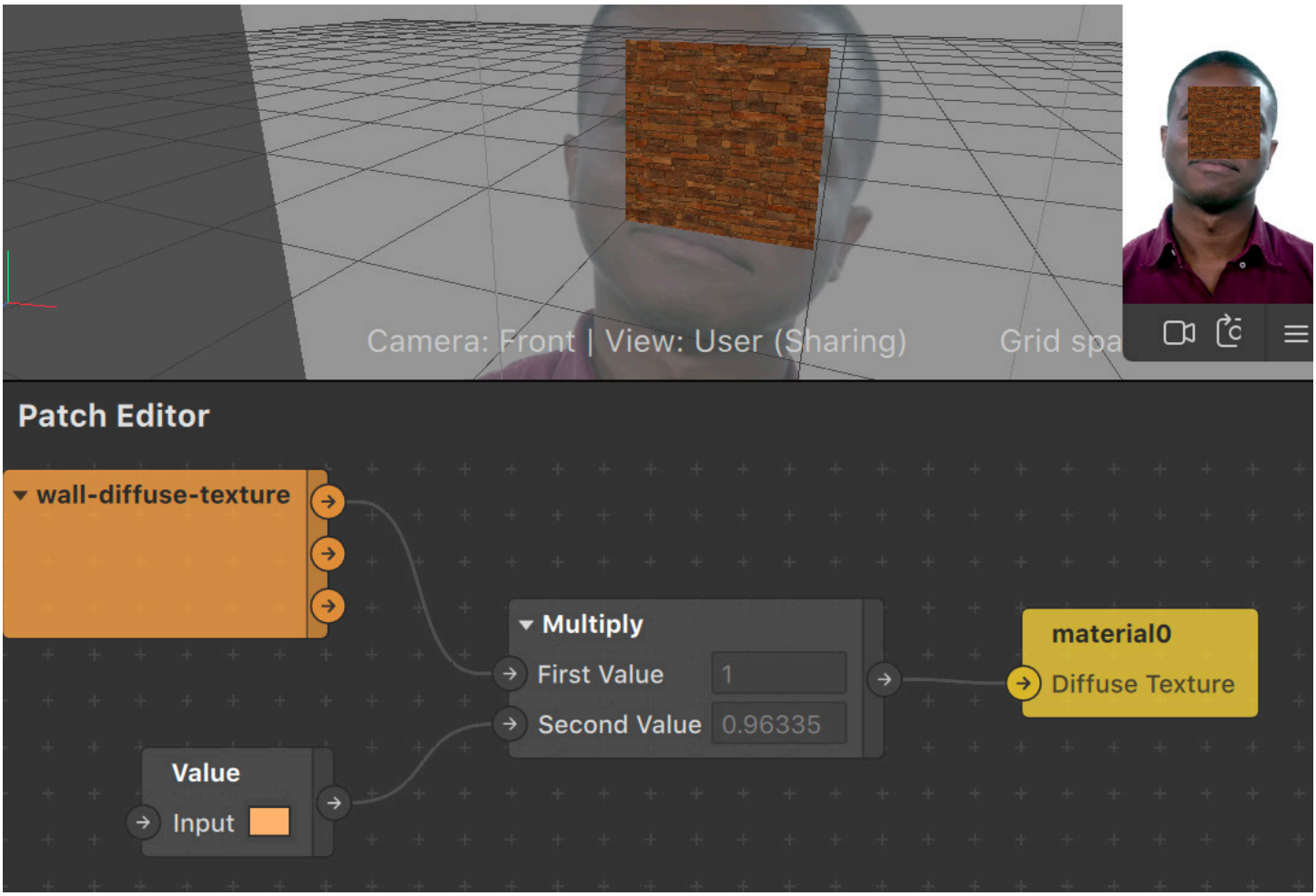
There are lots of types of patches to choose from in Spark AR Studio:

- Interaction – to interpret interaction from the person using the effect
- like screen taps and swipes.
- Face Landmarks – to precisely capture the position of facial features.
- Facial Gesture Patches – these patches detect facial expressions – like a 'happy face', 'kissing face' or 'surprised face'.
- Animation Patches – to add animation.
- Audio Patches – to manage and control your audio assets.
- Shaders – for working with textures and materials.
- Math – to perform mathematical functions.
- Logic Patches – to perform logical functions.
- Utility Patches – a broad range of patches, from face trackers to random number generators.
- User Interface Patches – to add UI functions to your effects.

What's a shader?

A shader is a set of algorithms that determines the appearance of a 3D object's surface. Shaders are particularly useful for applying complex changes to a material. Experienced creators can use Shader Code Asset to write custom shaders in Spark AR Studio.

You don't need to know how to code to create a shader. You can use the Patch Editor to create and connect a group of patches called a visual shader.



Creating a visual shader in the patch editor

Working with visual shaders

Visual shader patches can be used with other patches in a graph to change the appearance of a material. This is a way to apply more complicated changes than you could by using the shaders found in the Inspector.

Visual shaders can be used to determine the appearance of an object instead of applying Spark AR Studio's built-in shaders through the Inspector panel.

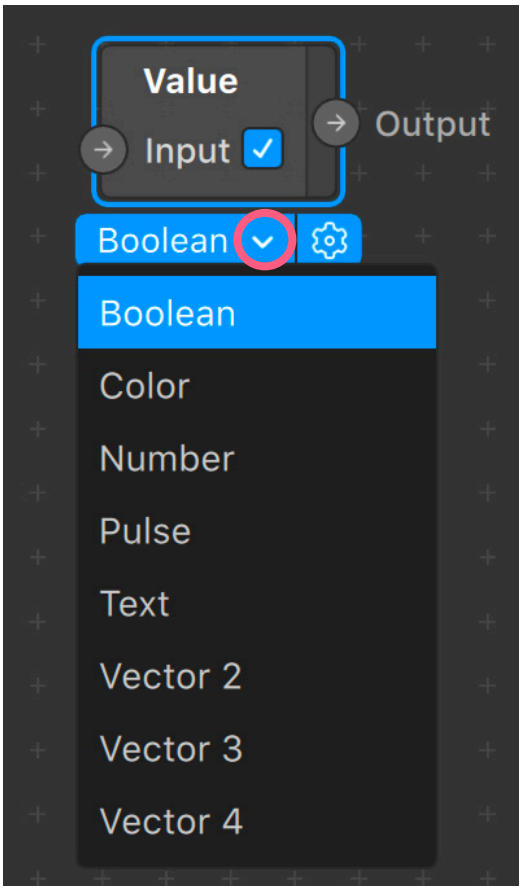
Like other patches in Spark AR Studio, there are a number of visual shader patches to choose from.

- Consumer Patch – visually represents a property of a material, like its position or scale. Usually appears at the end of a graph to specify how the material appears in the effect.
- Texture Asset – a producer patch that applies a texture to the material.
- Blend – blends two colors or textures together.
- Color Space – outputs a texture's RGB, HSV, and HSL color values. Useful for isolating the color values to change the texture's appearance.
- Composition – outputs two combined functions as a single data stream.
- Fallback – automatically outputs a fallback value when you disconnect the main value. For example, a color could be used as a fallback value for an image.
- Fragment Stage – processes each fragment in a texture individually instead of processing a texture at the vertices.
- Function – used as an identity function. It always returns the same value as its argument.
- Gradient – creates a grayscale gradient. Can be used with a Mix patch to create a color gradient.
- Gradient Step – defines each color you want to use in a color gradient. Use this patch multiple times to add multiple colors.
- Render Target Size – outputs the render target size. Useful for building effects using shaders requiring the exact pixel size, like pixel blurring.
- Shader Derivative – captures standard derivatives.
- Texture Sampler – samples a texture at the specified coordinates.
- Texture Transform – used with a 2D Transform Pack patch to scale, rotate, pivot and reposition a 2D texture on a 2D or 3D object.
- Vertex Attribute – provides the value of the selected vertex attribute among a selection of parameters.
- Vertex Transform – provides a vertex transformation matrix from a dropdown.

Working with data types

Every patch in Spark AR Studio has one or more ports. Connecting these ports to other patches will send and receive data.

You can change the data types of some ports. To do this, select the patch and click the dropdown in the menu.



Value patch data types

When you draw a connection from an output port Spark AR Studio will suggest patches that can be connected. If you try to connect ports with incompatible data types you'll see an error message.

Some data types are specific to Spark AR Studio. These ports can only be connected to other ports of the same type.

For example, the Face port in the face tracker patch outputs data on the position of the face. It can only be connected to another Face port, such as in a face landmark patch or an interaction patch.

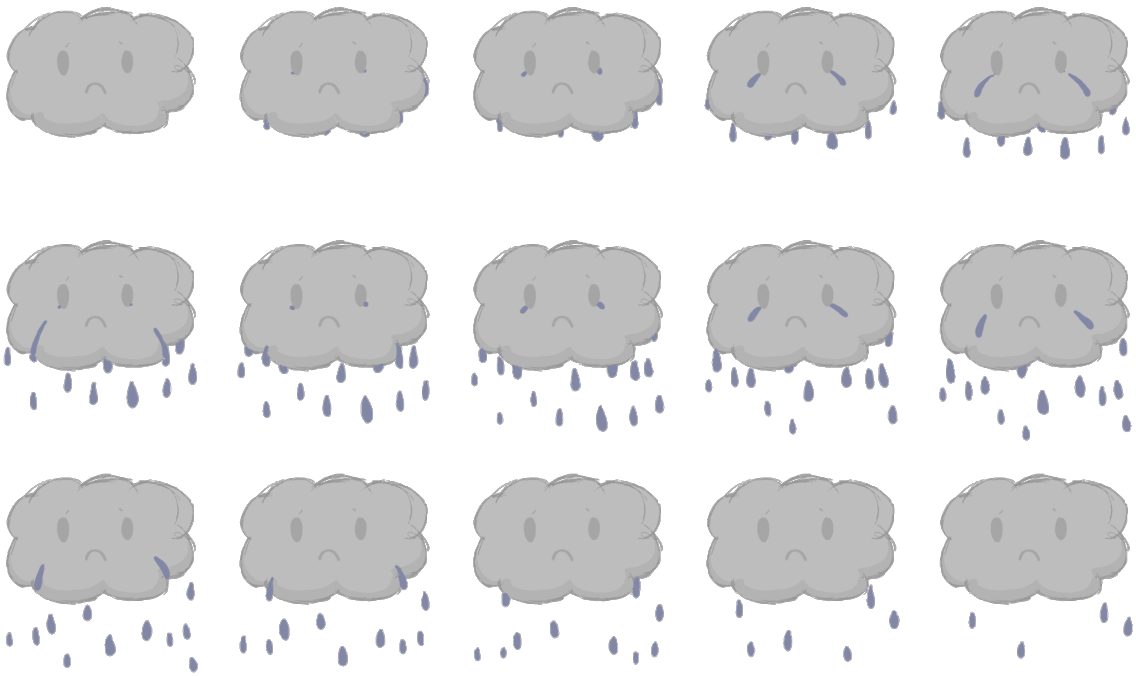
To learn more about how you can use patches to make your effect more interactive, visit Chapter 3: Adding Scene Interactivity.

You work with 2d and 3d objects to create effects, so it's important to understand what they are and how they function.

2D Objects

2D objects can be images, text, video or audio elements that are superimposed on the real world. For example, fonts, animations and sprite sheets are all 2D objects.

> Sprite sheets are single image files that contain multiple images within them, each a frame in a 2D animation.



Sprite sheet

3D Objects

When creating a Spark AR effect you'll be placing 3D objects within the real world so it's important that these objects represent a realistic and uniform scale. For example, if your effect allows people to place items from a furniture catalog into their living space, the 3D furniture should be realistically scaled.

There are 2 ways to enable **real world scale** in Spark AR Studio depending on the type of user experience you want to create:

Enable Real-World Scale in the Inspector

Your object appears realistically scaled but only after a real-scale signal is detected. This means the user sees the object abruptly change size at some point after opening the effect.

Enable Real-World Scale in the Inspector and add further logic via the Patch Editor or scripting.

This method allows you to hide the object until a real-scale signal is detected. This means the user only sees the object once it's realistically scaled.

Keep in mind that the number of triangles in a 3D object's mesh (or the 'triangle count) impacts performance, so it's a good idea to keep the triangle count as low as possible. The Spark AR Blender toolkit was developed with this in mind so that creators could optimize their 3D objects. The maximum number of triangles per object should be below 50,000. Keep the total triangle count for all objects in an effect below 150,000.

What makes a good target?

Not all images make good targets. Choosing a bad target image could mean that your effect won't respond to it.

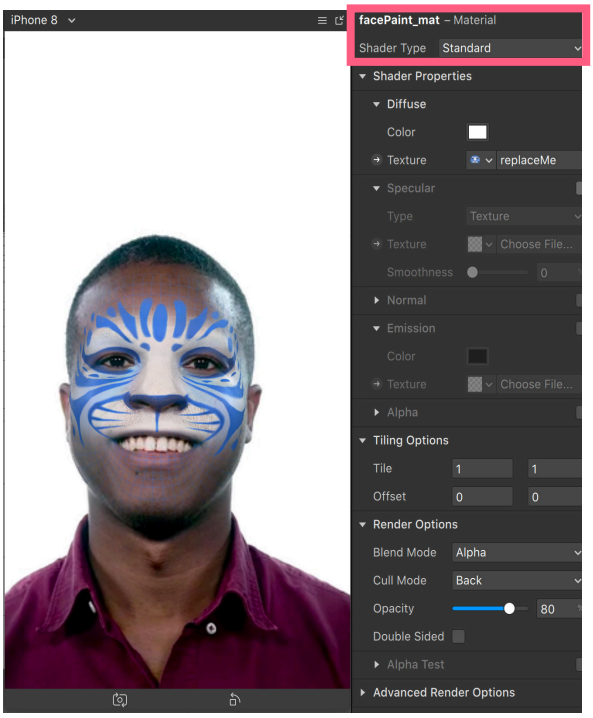
By making sure that your target image follows a few basic principles you can ensure that it gets detected by the camera.

- **Make it high contrast**. Both color and black and white images make for effective targets, as do all images with high tonal contrast. This means avoiding images with lots of pastel colors.
- **Make it high resolution**. An image should have a resolution of at least 300x300 pixels.
- **Make it sharp**. Try to stay away from smooth or soft edges and textures with a lot of gradients.
- **Make it asymmetrical**. Avoid symmetry and repeated patterns. Images that can easily be inverted might not track as well.
- **Make it complex**. Simple images might not work well. You should also avoid repeating the same motifs.
- **Make it flat**. Images on curved surfaces (like bottles) don't work well, as target tracking is meant for flat surface images.
- **Make it clear**. Target images can be all sorts of shapes, they don't have to be rectangular or square. However, you should still think about how clear the target image is.
- **Avoid blank space**. Avoid targets that have a lot of transparency or blank space around the actual image
- **Focus on placement**. Target AR effects work best when the camera is closer to the image, so it fills a large portion of the device frame. Large targets work better too. Small images like stickers might not work, even when in the full frame.

CHOOSING WHICH ASSET IS THE MOST APPROPRIATE FOR YOUR DESIGN.

Working with materials

Materials are always added to an object to help bring it to life. Here we break down one-by-one the four materials that you'll primarily be working with in Spark AR.



Standard Material

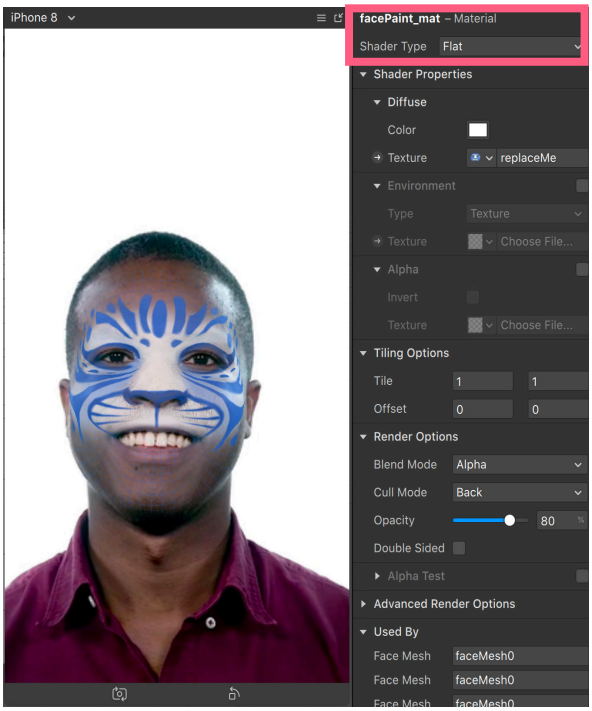
Standard Material

The standard material adds realistic depth to 3D objects, using a lighting system called the Phong model.

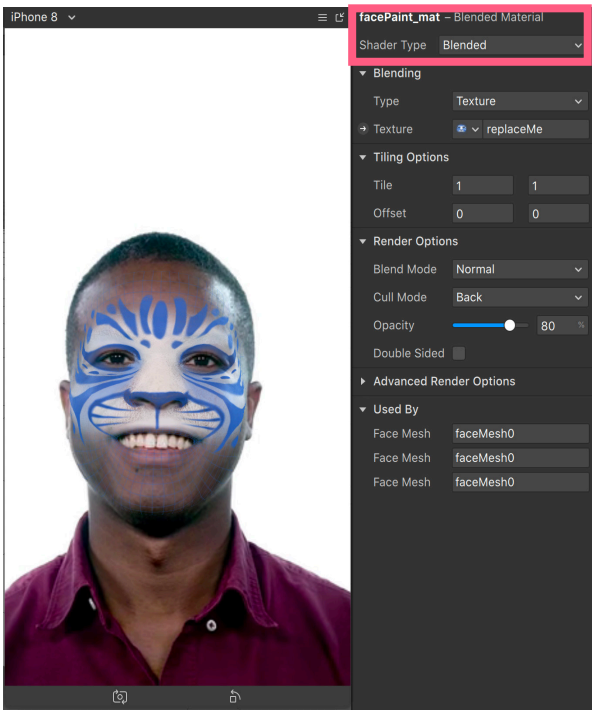
When you create a material in Spark AR Studio it will generally be a standard material, unless you create it for a 2D object.

Properties of a Standard Material:

- **Diffuse** - define the base color and pattern of an object. You can use Color and Texture together to create variations in the texture map.
- **Specular** - define the shininess and highlight color of a surface.
- **Normal** - create realistic textures like bumps, grooves and rivets without adding extra geometry to your object.
- **Emission** - allow your material to illuminate from itself. It won't cast rays, so the illumination won't affect other objects in the scene.
- **Alpha** - mask the alpha channel. Use Invert to switch which part of the texture is masked.
- **Tile** - Scale the textures you've applied to your material.
- **Offset** - Shift the origin of your textures.
- **Render Options** and **Advance Render Options** - Edit these properties to control how the material renders in the scene.
- **Used by** - The object this material is applied to.



Flat Material



Blended Material

Flat Material

Flat materials don't respond to lighting and display color and texture values absolutely.

You might use this material if there's lighting and shadow present in your texture file already or if you're creating a material for a 2D object.

Properties of Flat Material:

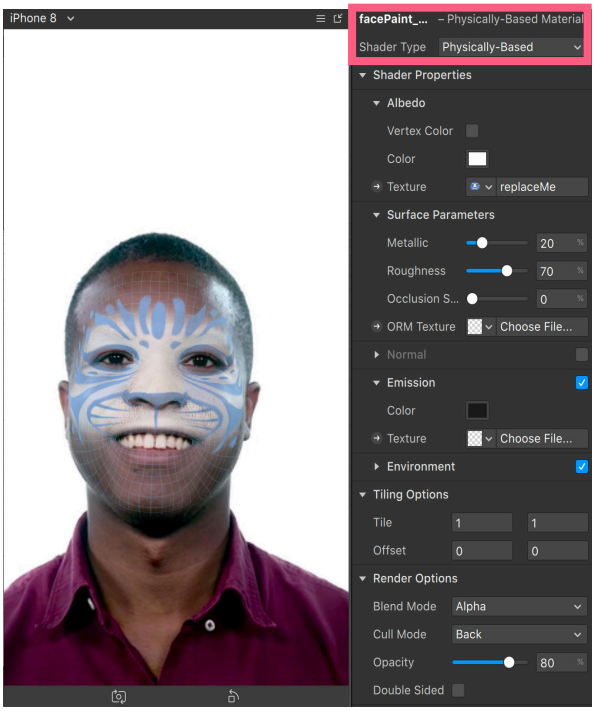
- **Diffuse** - edit the Color option to add a color from the palette to your material.
- **Environment** - check this box to add an environment texture to your material.
- **Alpha** - use this option to mask the alpha channel. Use Invert to switch which part of the texture is masked.
- **Tile** - scale the textures you've applied to your material.
- **Offset** - shift the origin of your textures.
- **Render Options** and **Advance Render Options** - edit these properties to control how the material renders in the scene.
- **Used by** - the object this material is applied to.

Blended Material

The blended material includes a blending mode, to mix textures and colors together.

Properties of the Blended Material:

- **Texture** - choose the texture file you want to add from the menu or import a new one.
- **Color** - add a color from the palette.
- **Blend Mode** - sets the blend mode of the material.
  - Normal
  - Add
  - Multiply
- **Opacity** - set the transparency of the material on an individual map level.
- **Double-sided** - displays both sides of a face normal.
- **Used by** – check which object this material is applied to.
- **Cull Mode** – describes how material is culled:
  - Front – discard pieces of the mesh that are facing the viewer.
  - Back – discard pieces of the mesh that are facing away from the viewer. Most effects use this mode to improve performance.



Physically-Based Material

The Physically–Based Material (PBR)

This material is used to create realistic-looking objects in Spark AR Studio. The different properties in this material allow you to add surface roughness, metallic effects, and lighting that mimics real world light.

For a complete physically-based material, you need:

- **A base texture** – this forms the base of your material. Use it to add color and details.
- **An ORM texture** – this allows you to define the roughness, metallic details and occlusion strength of the material.
- **A Normal map** – to create the appearance of real-world texture like bumps, grooves and rivets without adding extra geometry to your object.

When you've created a physically-based material you'll be able to edit these properties in the Inspector:

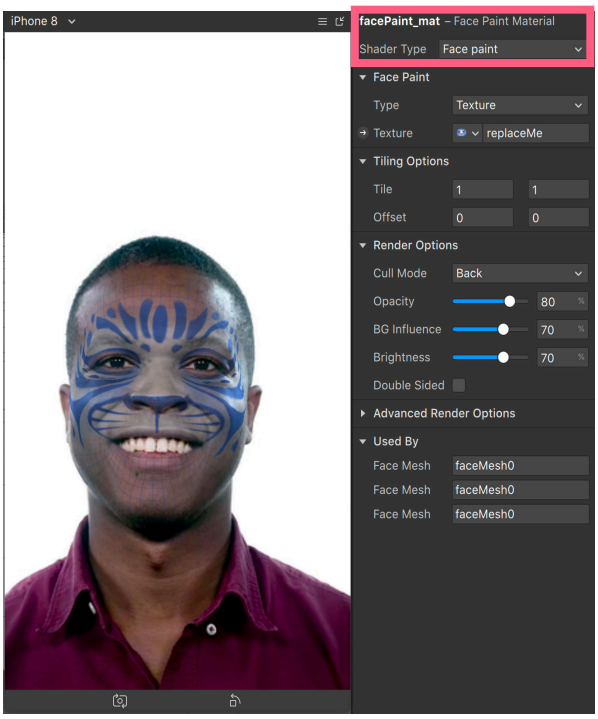
- **Albedo** – this is the base color and texture of the material. It offers two options:
  - **Color** – use this option to choose the base color of your material.
  - **Texture** – apply a base texture. The texture will be modulated by the color you've selected.

The Surface parameters dropdown is where you apply an ORM texture. You can adjust the slider to control:

- **Occlusion** – this option approximates soft shadows baked into the creased areas of a surface.
- **Roughness** – to add rough surface details.
- **Metallic** – to add metallic surface details.

Three tips for creating materials

1. Use Physically-Based materials (PBR) and Environment Textures.
2. Use as few materials as possible.
3. Use the Face Mask Template in Adobe® Photoshop® to create face assets.



Face Painting

The Face Paint Material

The face paint material is used to create a mask effect that shows the skin and features of the person using the effect, either behind a custom texture or a color. For example, you can use this material to create a tattoo or makeup effect.

Properties of the Face Paint Material:

- **Type**
  - Texture - choose the texture file you want to add from the menu or import a new one.
  - Color- add a color from the palette.
- **Tiling Options**
  - Tile
    - Scale the textures you've applied to your material.
  - Offset
    - Shift the origin of your textures.
- Render options and advanced render options
  - Edit these properties to control how the material renders in the scene.
- Used by
  - The object this material is applied to.

Three tips for creating materials

1. Use as few materials as possible.
2. Use the Face Mask Template in Adobe® Photoshop® to create face assets.
3. Use Physically-Based materials (PBR) and Environment Textures.

The Blend Mode enum describes how material is blended. Values:

- **Subtract** – subtracts pixel values of the source from the destination. In case of negative values, black is displayed.
- **Replace** – the non-transparent pixels of the source replace corresponding pixels of the destination.
- **Alpha** – linear interpolation between the pixels of the source and destination as specified by the alpha values of the source.
- **Associate_Alpha** – same as alpha blend mode, but for source with RGB channels of each pixel premultiplied by its alpha.
- **Add** – adds pixel values of the source to the destination.
- **Multiply** – multiplies the RGB channels of each pixel from the source with the values of the corresponding pixel from the destination.
- **Screen** – values of each pixel in the source and destination are inverted, multiplied, and then inverted again.

Skin retouching

Skin retouching is an effect that can be applied when we want to give a smoother appearance to a face or scene, using a material in Spark AR Studio.

Working with textures

A texture is an image file used to help define the appearance of an object in your scene.

You can apply one or more textures to a material, then apply the material to an object.

Texturing with realistic lighting is difficult to achieve in AR effects and it takes a lot of processing power. To reduce the impact on performance you can do a few things:

1. Bake lighting directly into 3D objects used for decoration in face effects to hint at highlights and shadows.
2. Bake lighting into static (non-moving) objects such as furniture and accessories for a more realistic effect.
3. Use a soft light setup without strong highlights or shadows. Baked lighting works best in environments where the light, intensity, and color don't change throughout the effect.
4. Use MatCap (material capture) reflection environment textures with flat shaders.

You're able to make a variety of textures in Spark AR Studio. For example, you can turn the video captured by the camera or the face tracker into a texture. You can also use visual shaders to create color gradients, and signed distance fields (SDF) to create shapes and patterns. You can even use the segmentation texture to separate the user from their background to transform the environment behind them.

Compressing textures can make your effects perform better. It's a good idea to look for opportunities to compress textures in your effects where you can.

A few things to consider when working with textures:
Spark AR Studio supports PNG and JPG texture files. Files can be a maximum of 1024 x 1024 pixels in size.

To achieve interesting yet efficient texture results:

1. Use textures in sRGB space with a maximum resolution of 1024 x 1024 pixels.
2. Use Spark AR templates to make your textures compatible if you've created them using Adobe® Substance Painter® texturing software.
3. Merge materials as much as possible for performance benefits.
4. For textures with a repeating pattern, import a lower resolution image and use Spark AR Studio's visual shaders and texture tiling feature instead of using a larger resolution image.

What about file size for textures?

Spark AR studios supports both PNG and JPG texture files, but to achieve efficient results you should use textures in sRGB space with a maximum resolution of 1024 x 1024 pixels.

Pro tip: When transparency isn't needed, use lower resolution JPG files instead of PNG files.

If you create your textures with Adobe® Substance Painter® texturing software use Spark AR templates to make your textures compatible. Delete any unused textures from the project in order to save space.

Merge materials as much as possible for performance benefits. For textures with a repeating pattern, import a lower resolution image and use Spark AR Studio's visual shaders and texture tiling feature instead of using a larger resolution image.

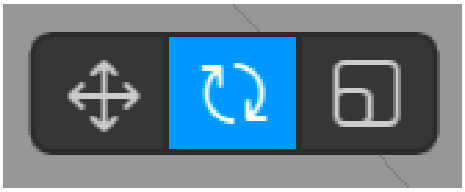
How to integrate assets into a scene

How you add an asset to your scene, whether by using patch editor or canvas, depends on the type of asset it is. For example:

- Textures, including animated textures, are applied to material. You then apply the material to an object in the scene using the Material option in the Inspector.
- Audio files should be connected to an asset called an audio playback controller, which can then be connected to an object called a speaker.
- 3D objects can be dragged from the assets panel to the Viewport or Scene Panel.

If a 3D object includes an animation, you can add the animation using the Animation property of the 3D object.

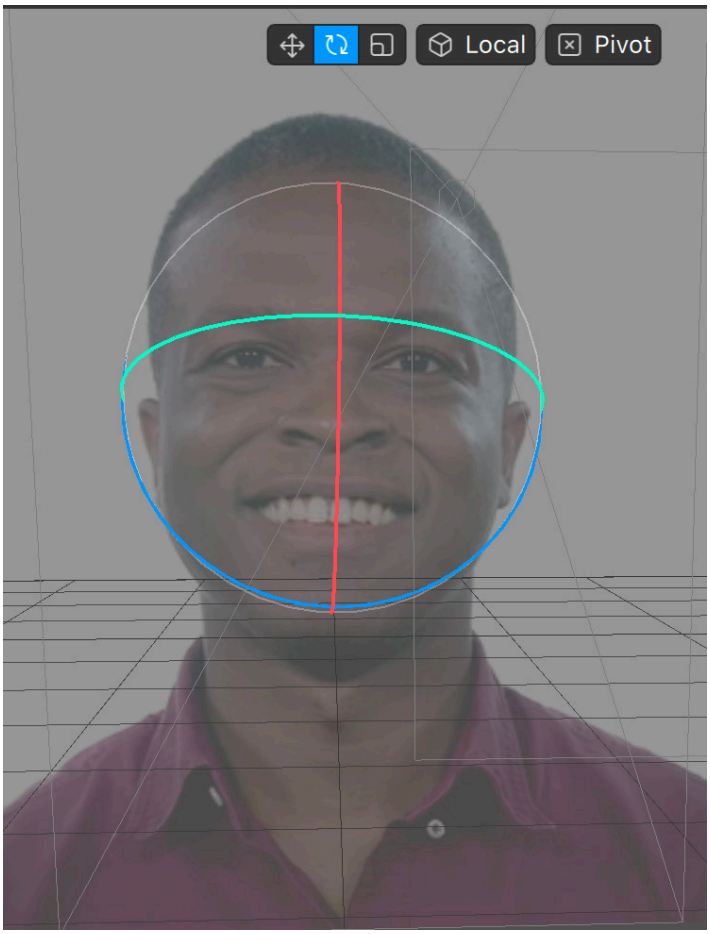
You can import all kinds of assets into Spark AR Studio – for example, textures you've made in external programs and audio files.



Use the Manipulators when you want to move large objects.

The Manipulators are the 3 icons at the top of the Viewport. You can use the Manipulators to quickly change your object's:

- **Position** – to choose where your object is placed within the scene
- **Scale** – to make your object appear bigger or smaller
- **Rotation** – to rotate your object



Manipulators interface in the viewport

These are known as transformations. You can also edit these properties in the Inspector.

When an object is a child of another object, it can be transformed in relation to its parent object's coordinates, which might not be the same as the Global coordinates — this is called Local. Or that same object can be transformed based on the fixed X, Y and Z axis — this is called Global. Users have the option to make a patch for the object in the scene by clicking 'create' which is located next to 'patch' in interactions.

For bigger changes you can use the simulator as it allows you to drag entire objects across the screen.

But if what you want is to make small changes to an object, the properties panel is the best place to do so. There you can make minor tweaks to your object by inputting any number you like until you land on your preferred scale, position or rotation.

Integrating audio assets into your effect

There are lots of options available for adding audio to your Spark AR Studio effects.

All Spark AR Studio effects include a microphone. This captures the voice of the user and plays it back into the effect. You can disable the microphone if you want to have an audio file playing in your effect on load instead of capturing the sound from the device's microphone. An object called a speaker can be added to render ambient and one-shot audio clips in the scene.

To connect an audio clip to the speaker, you can use an audio playback controller or use the audio-player and single or multi-clip controller patches.

The audio playback controller lets you play an audio clip on a loop in the background of your effect or play a one-shot sound in response to a boolean signal. While the Audio-Player and Single or Multi-Clip Controller patches are used for more complex playback systems — for example to modify the audio live through pitch shifting or distortion.

Spark AR Studio also includes a wide range of audio effect patches that you can use to modify audio clips or the sound picked up by the microphone.

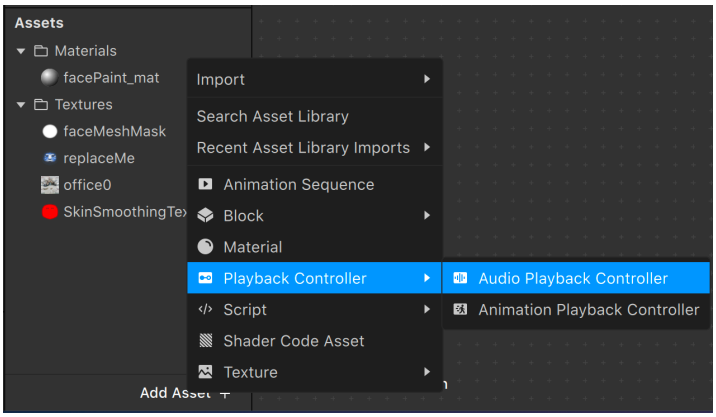
Tip: To use the audio playback controller, search for it under the 'add assets' tab in the assets window.

Before creating your ambient audio asset you should consider:

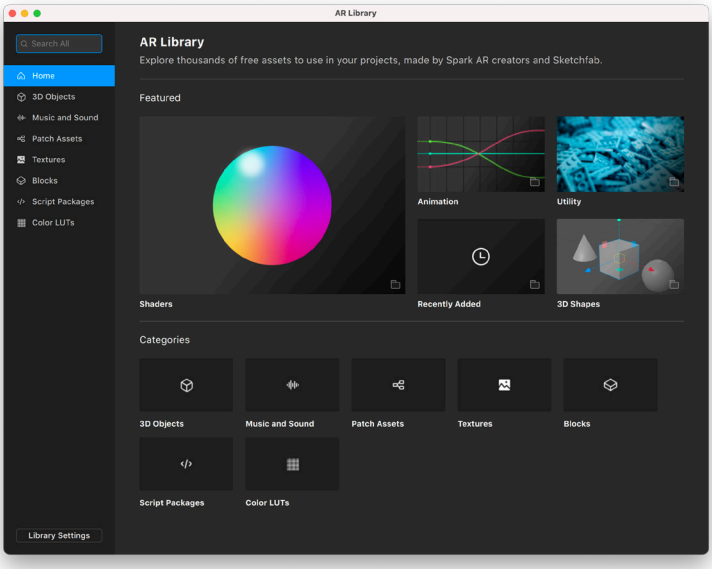
1. What you want people to use them for.
2. How your effect sounds on a mobile device.
3. Whether you want the file to loop seamlessly or not (we recommend a small, looping ambient bed).
4. If you choose not to use a seamless looping ambient bed, make sure that your ambient bed runs the max capture length, which for the Facebook camera is 20 seconds.

Using AR music for Instagram

To create an effect that responds to music, you can either import and use your own legally licensed music files or you can let people pick a song they like from the thousands of free tracks in Instagram's music library. This feature is now available for Android and iOS users.



Audio Playback Controller in the assets panel



AR library

Adding 3D objects to your Spark AR effects is a great way to create more immersive effects that take place in people's real surroundings.

You can create your own 3D objects with tools such as Blender, a free and open-source 3D content creation suite.

You can also create your own 2D assets via third-party software integrations such as Adobe® Substance Painter®, Adobe® Photoshop® and Photopea. To learn more about third-party asset creation tools by visiting Spark AR Learn.

The AR Library and in-app integration

The AR Library is used to import complete 3D objects, audio clips, patch assets, textures and scripts into your Spark AR Studio projects. It provides a quick and easy way to start building your own effects.

The AR Library includes the following assets provided by Facebook and Sketchfab:

- 3D objects and patch assets are licensed for use under Spark AR Studio Terms as well as the CC BY 4.0 license.
- Scripts are licensed under the MIT license.
- Audio files are licensed under Facebook's Sound Collection Terms.

Spark AR Toolkit is an add-on that you can use in Blender to optimize your 3D object and check that it's ready to import to Spark AR Studio.

Use Spark AR Toolkit to:

- Make your 3D objects more lightweight.
- Set the height and the pivot point position.
- Export quickly and easily

Chapter 2

FEATURES OF SPARK AR STUDIO



Now that you understand all of the elements and components of Spark AR, it’s time to learn how to create an interactive scene using the effects you’ve created.

UNDERSTANDING AND APPLYING CONCEPTS

Scripting with Javascript

Spark AR Studio supports JavaScript for adding logic and interactivity to your effects. Many capabilities available in JavaScript can be added to your effect as patches, so you can add interaction and complexity without knowing how to code.

Scripting is broken down into modules with each module implementing a particular functionality. For a complete list of modules, each with their own code example, check out the Scripting Object.

Spark AR studio opens your scripts with the default editor assigned to JavaScript/TypeScript files on your macOS or Windows operating system. If you don't already have one installed, you can download an editor such as Visual Studio Code and set it as the default app for opening JavaScript/TypeScript files.

Be aware when scripting in Spark AR Studio that:

- JavaScript ES6 and TypeScript are both supported.
- You can split scripts across multiple files.

The Javascript patch bridge

You can pass information between scripts and the Patch Editor by adding shared variables. Shared variables can be created from the Inspector Panel after selecting a script file in the Assets Panel. There are 7 variable types supported with some type differences between the Patch Editor and script:

Patch Editor type	Script type
Boolean	Boolean
Number	Scalar
Pulse	EventSource
Text	String
Vector 2	PixelPointSignal
Vector 3	VectorSignal or PointSignal
Vector 4	RgbaSignal

Should you use Patch Editor or Scripting?

Scripting is more complex but it's also a more time efficient option if you already know how to code. Scripting allows you the flexibility to run loops/ iterations and save variables, whereas patches have to be manually connected one by one. You can't go wrong with either option, which one you use depends on your preference and experience level.

Reactive Programming in Spark AR Studio

Spark AR Studio's implementation of reactive programming allows you to create relationships between objects, assets and values. This means that the engine doesn't have to execute JavaScript code every frame when performing common tasks such as animating content, looking for user input, or realigning a mask to a face.

Reactive programming is also compatible with visual programming, reducing the frequency of calls made into the scripting engine. The reactive programming model allows you to define relationships between objects and values in such a way that a value bound to a signal is automatically updated when the latter changes.

If you've worked with other development tools in the past you may be more familiar with imperative programming. This model differs from reactive programming in that statements are executed in order and values are set only when an expression is evaluated. With a reactive model the updating of the value happens in native code rather than JavaScript, reducing the impact on application performance. The reactive model's compatibility with visual programming also means that the frequency with which calls are made to the scripting engine is reduced. Both this and the previous points result in performance improvements for your effects.

How to use variables, logic and signals

What is Logic?

Logic patches define the conditions needed for something to happen in an effect. For example, you can use logic to trigger an animation to play when the user interacts with the device or after an effect runs for a certain number of seconds.

Patch	Description
And	Make something happen when 2 conditions are met at the same time, like when the user opens their mouth while raising their eyebrows.
Equals	Make something happen when the value coming from the input patch is within a close range or around a certain number. For example, if you want the user to face roughly a certain point on an axis to make something happen.
Equals exactly	Make something happen when the value coming from the input patch is an exact number.

Patch	Description
Greater Than	Make something happen when the value coming from the input patch is greater than a preset value.
Greater or Equal	Make something happen when the value coming from the input patch is greater than or equal to a preset value.
If Then Else	Switch between two states in response to something depending on the data type selected.
Less or Equal	Make something happen when the value coming from the input patch is less than or equal to a preset value.
Less Than	Make something happen when the value coming from the input patch is less than a preset value.
Not	Reverse the output of a connected patch.
Or	Make something happen when one of 2 conditions are met, like a tap or a long press on the device screen.

A signal is a special object containing a value that changes over time.
One of the ways Spark AR Studio makes use of a reactive programming model is by allowing you to treat values as signals.

There are signal equivalents for primitive data types including number (ScalarSignal), string (StringSignal) and boolean (BoolSignal) values. The syntax for binding a signal to an object is the same as performing a standard assignment.

When a number, string or boolean value is passed to a function or property setter that expects a signal, the value is implicitly converted to a constant signal. You can also explicitly convert a primitive data type to its signal equivalent with the val() method exposed by the ReactiveModule class.

If you need to retrieve the value that a signal contains you can use the pinLastValue() method exposed by the ScalarSignal, StringSignal and BoolSignal classes.

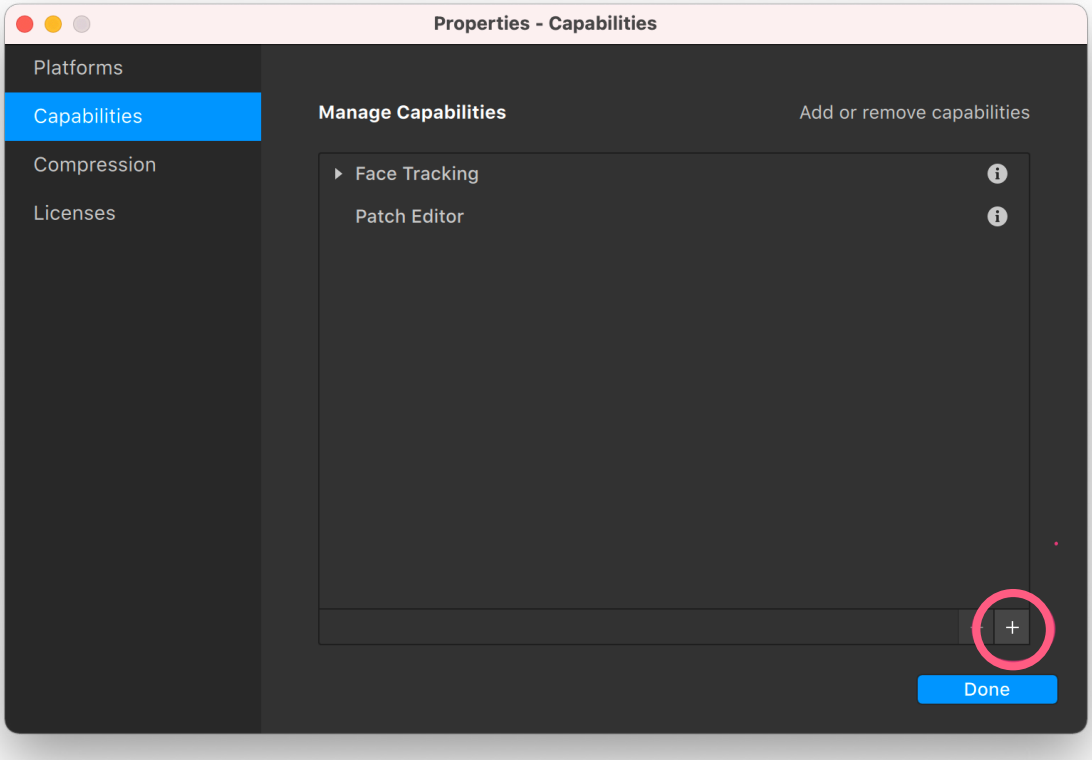
As their value changes over time, standard JavaScript operators such as +, -, * and / can't be performed directly on signals. Instead, these types of calculations are performed via methods exposed by the ReactiveModule class and the ScalarSignal class.

The watch() method exposed by the DiagnosticsModule class allows you to add a signal to the watch view in Spark AR Studio to monitor how its value changes over time.

ASSIGNING AND CONFIGURING ASSET PROPERTIES

Configuring settings for a project

Not all capabilities are added to projects automatically. If needed, other capabilities may need to be added manually. You can configure the capability to be used via the Project Tab at Spark AR Studio > edit properties... > capabilities > + to add new capabilities.



Project properties

You can also set the type of the experience for the effect and the compression configurations.

While we do launch new capabilities often on both platforms, keep in mind that there are some capabilities that are platform-specific. Being aware of these limitations will help you manage client expectations.

Camera configuration and visibility

The camera is an object that's automatically listed in the Scene panel in every effect. It represents the camera in the device showing your effect, which means it can't be removed from a project.

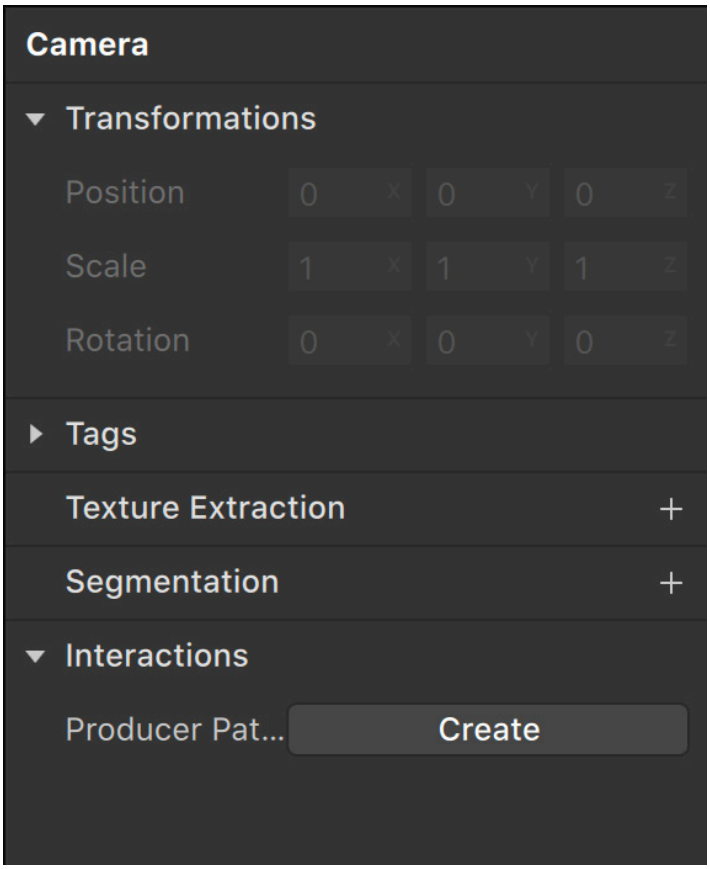
The camera is listed at the top of the Scene panel. It's labeled Camera. You can use the Camera to add background segmentation or create a texture representing the video captured by the camera while your effect is being played.

When objects are children of the camera in the Scene panel, they move with the movement of the camera.

When objects are no longer children of the camera you can position them in a fixed place in the world, creating a world effect.

When the Camera is selected in the Scene panel, you can see its properties in the Inspector. The following are properties of the Camera:

- **Transformations** – the Position, Scale and Rotation can't be edited.
- **Texture Extraction** – click the arrow next to Texture Extraction to create a texture from the video captured by the camera.
- **The texture** is listed in the Assets panel as cameraTexture0.
- **Segmentation** – segmentation separates part of the scene so you can make changes to it separately from the rest of the effect.



Camera properties in the inspector panel

What is segmentation and how does it function in AR?

Segmentation is used in Spark AR Studio to detect and separate parts of the camera input. For example, to separate someone's hair from the rest of the scene, which would allow you to change the color without changing anything else in the effect.

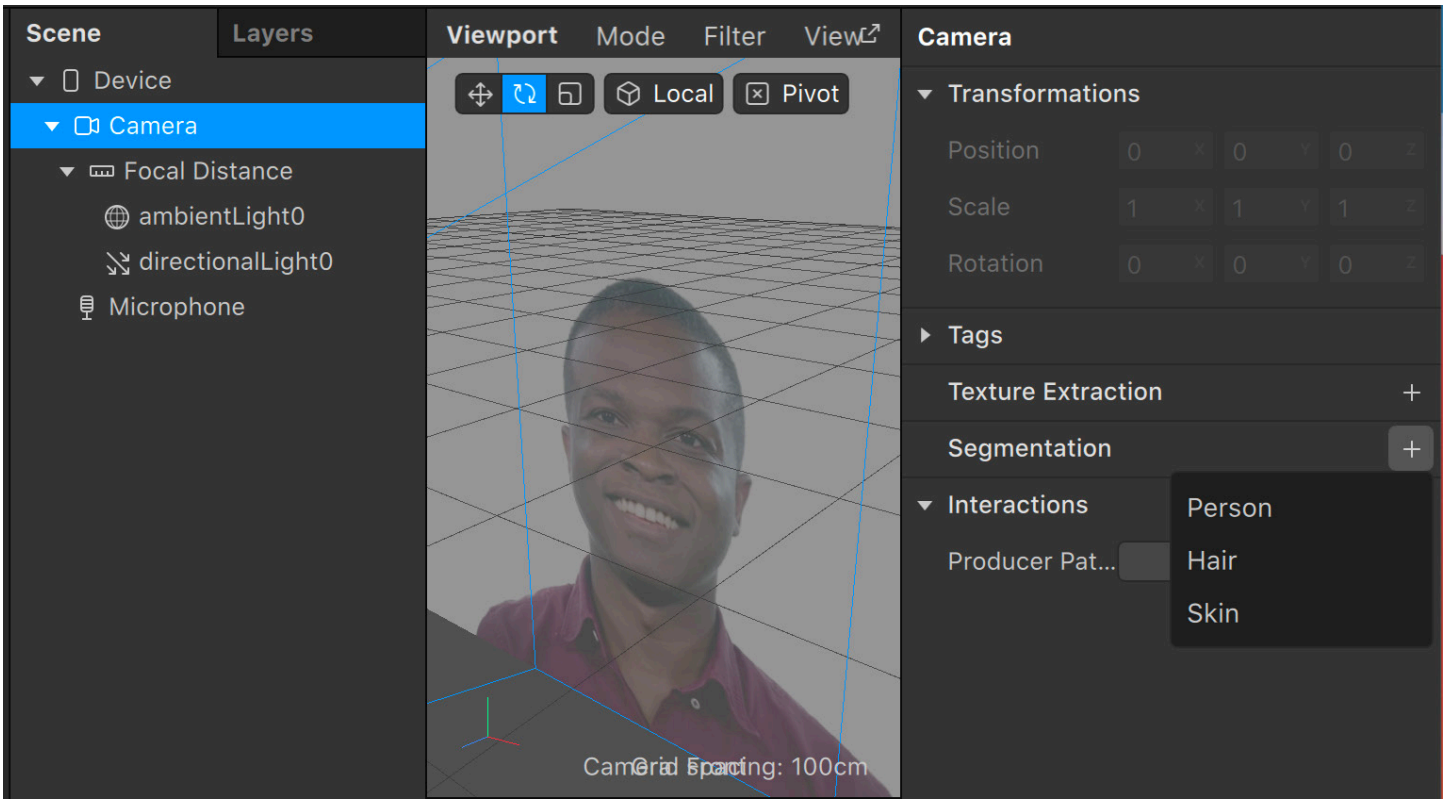
You can add segmentation to an effect by applying a 'segmentation texture' to the Texture property of a material. You would then apply this material to a rectangle that covers the full width and height of the device screen.

There are 3 types of segmentation available, and you can combine all 3 types of segmentation to achieve a more complex effect.

Person segmentation
to separate the person using the effect from their background.

Hair segmentation
to separate someone's hair.

Skin segmentation
to separate any skin detected by the camera from the rest of the scene.



Segmentation

ASSIGNING RELATIONSHIPS BETWEEN OBJECTS

Using the Scene Tree in the Studio to create relationships

You can create relationships between objects in the Scene panel. Creating a child-parent relationship means you can make the same change to multiple objects at the same time, by making a change to the parent object. To create a child-parent relationship, drag the object that you want to be the child onto the object that you want to be the parent.

This is also where you'd insert an element that makes your effect respond to the person using it, or someone's environment. For example a face tracker if you want to build an effect that responds to someone's face.

Grouping with Null Objects

In Spark AR Studio a null object can be used to group other objects. It doesn't have a shape, but it can be manipulated in 3D space.

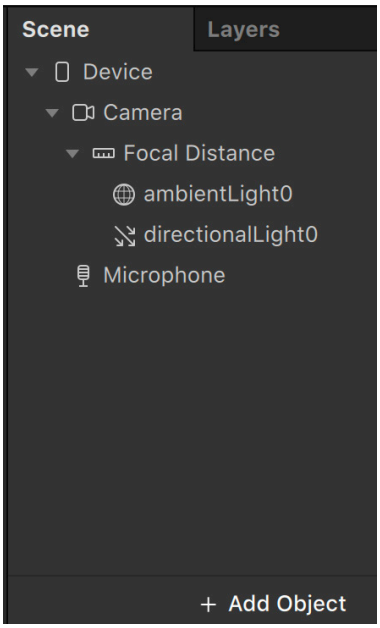
Objects grouped under a null object are known as its children. The children take on the properties of the null object. For example, if you make a null object bigger, any children will get bigger too.

Null objects are particularly useful for adding secondary animation or to make sure a group of objects have exactly the same position.

Grouping other objects

To group objects using a null object, the objects must be children of the null object.

You can also use patch logic to assign relationships between objects. Here's how:



The scene panel

- If you click the arrow next to a property in the Inspector, it creates a patch to represent it.
- Then **you can choose between 2 interactions**:
  - **Object Tap** – to create a patch representing the null object (and its children) in the scene and an Object Tap patch, which will detect when someone taps the null object.
  - **Producer Patch** – to create a patch that represents the null object.

Deleting objects

You can delete unused assets by checking the 'Used By' property of the asset, which states whether it's related to another asset in the project file. If empty, consider it safe to delete. Child-parent relationships between grouped objects can be removed by dragging and moving the child object outside the parent object.

Null objects can have certain properties:

- **Visible** – clear this box to stop the null object and any children from being rendered in the scene.
- **Transformations** – change the position, scale and rotation of the null object. You also use the manipulators to do this.
- **Interactions** – insert simple combinations of patches involving the instance into the Patch Editor, like tap gestures.
- **Enable For** – choose the camera or cameras on a mobile device in which you want to render the null object.

Rendering and layering objects

Objects in a Spark AR Studio effect are always rendered in a specific order. In general, an object that's listed closer to the Camera in the Scene panel appears in front of one that's further away. This can mean an object intended to appear in front of another object appears behind it, and vice versa. You can use layering to change this. By default, new projects have one layer which every object you add the scene is assigned to. There must always be one layer in a project.

The render order is the sequence in which the layers are rendered. If you have more than one layer in your project, the layer at the bottom of the list is rendered first and the layer at the top of the list is rendered last.

Opaque objects are only affected by the render order if their position on the Z-axis overlaps. Transparent and semi-transparent objects may be affected by render order regardless of their position on the Z axis, because they can occlude objects in other layers. This means transparent objects set to render first will occlude objects rendered on a subsequent layer. You might need to adjust your layers to account for this.

Render options

You can use the render options and advanced render options in premade materials to set how the material renders in your Spark AR effect.

Render options and advanced render options can be edited in the Inspector, when a material is selected in the Assets panel.

- **Cull Mode**
  - Back – discard pieces of the mesh that are facing away from the viewer. Most effects us this mode to improve performance.
  - Front – discard pieces of the mesh that are facing the viewer.
- **Blend Mode**
  - Sets the blend mode of the material.
- **Opacity**
  - Sets the transparency of the material on an individual map level.
- **BG Influence**
  - Multiplies a color or texture against a greyscale version of the video feed. Only available in the face paint material.
- **Brightness**
  - Sets the brightness of a color. Only available in the face paint material.
- **Double-Sided**
  - Displays both sides of a face norm
- **Alpha Test**
  - Use the slider to make semi-transparent parts of a material transparent.

There are also several advanced render options

- **Use Depth Test and Write to Depth Buffer**
  Use the Write to Depth Buffer and Use Depth Test boxes to control certain order parameters for the object the material is applied to.
- **Both boxes selected**
  Enable read and write depth buffer. This means that closer objects will obscure far objects.
- **Write to Depth Buffer box selected**
  Disable read depth and enable write depth. These objects will always draw over any other object. They will obscure parent objects that are drawn after unless these objects have either the Write to Depth Buffer or both boxes selected.
- **Use Depth Test selected**
  Enable read depth and disable write depth. These objects will be obscured by nearer objects that have already been drawn that have either the Write to Depth Buffer or both boxes selected. These objects can be obscured by any objects drawn afterwards, whether they're nearer or further away.
- **Clear both boxes**
  Disable read and write depth. These draw on top of any objects of any type already drawn, and in turn will be overdrawn by any later rendered objects, regardless of whether they're nearer or further away.
- **Color Mask**
  Use these boxes to disable writing to specific channels of the color buffer. The color channels of the cleared editors will be preserved after rendering. When using this feature, the color in the Simulator and the viewport might not be the same. The Simulator will most closely match the end result on a device.

Chapter 3

ADDING SCENE INTERACTIVITY



Being able to interact with an effect and becoming part of a scene is what makes AR such an exciting technology. Once you've integrated objects into your scene, the next step is to make it engaging for users so that they'll want to share it with friends and continue using it themselves.

PROCESSING SIGNALS WITHIN A SCENE

HOW TO IMPLEMENT LOGIC AND USER INPUT WITH PATCHES

You can use utility patches for a variety of functions, such as controlling outputs and transforming data types.

The Animation patch can be used to drive an animation triggered by an Interaction patch or logic that plays once.

Frame Transition patch can be used to control the frames in an animation sequence.

To drive animations that play on a loop, use the Loop Animation patch.

Interaction patches in Spark AR Studio detect people's taps, swipes and gestures on the screen of their device and add interactivity to your effects in response.

It's a good idea to add this type of interactivity to effects that use the plane tracker to place objects on surfaces in the real world. That way people can make objects the right size and move them around.

Some of the patches you can use to make your effect more interactive include:

- The **Screen Pan patch** to control position
  - The Screen Pan patch detects a swipe of the finger across the
  - device screen.
- The **Divide patch** to manipulate the input from the Screen Pan patch.
- The **Screen Pinch patch** to change scale.
  - Adding these patches to your effect will make an object get bigger or smaller when someone pinches the screen.
- The **Screen Rotate patch** to rotate an object.

Tap Patches

- The **Screen Tap patch** is used to make something happen when the screen is tapped.
- The **Object Tap patch** is used to make something happen in your effect when someone taps a specific object.

Here are some of the interaction patches you can use to add movements and gestures to your effect:

Movements

Patch	Description
Blink	Make something happen when the user blinks.
Eyebrows Lowered	Make something happen while the user's eyebrows are lowered.
Eyebrows Raised	Make something happen while the user's eyebrows are raised.
Head Nod	Make something happen when the user nods.
Head Rotation	Make something happen while the user's head moves in a certain direction.

Patch	Description
Head Shake	Make something happen when the user moves their head from left to right or right to left.
Left Eye Closed	Make something happen while the user's left eye is closed.
Mouth Open	Make something happen while the user's mouth is open.
Right Eye Closed	Make something happen while the user's right eye is closed.

Gestures

Patch	Description
Happy Face	Make something happen while the user makes a happy face.
Kissing Face	Make something happen while the user brings the corners of their mouth together to make a kissing face.
Smile	Make something happen while the user smiles.
Surprised Face	Make something happen while the user makes a surprised face.

The Face Landmark patch

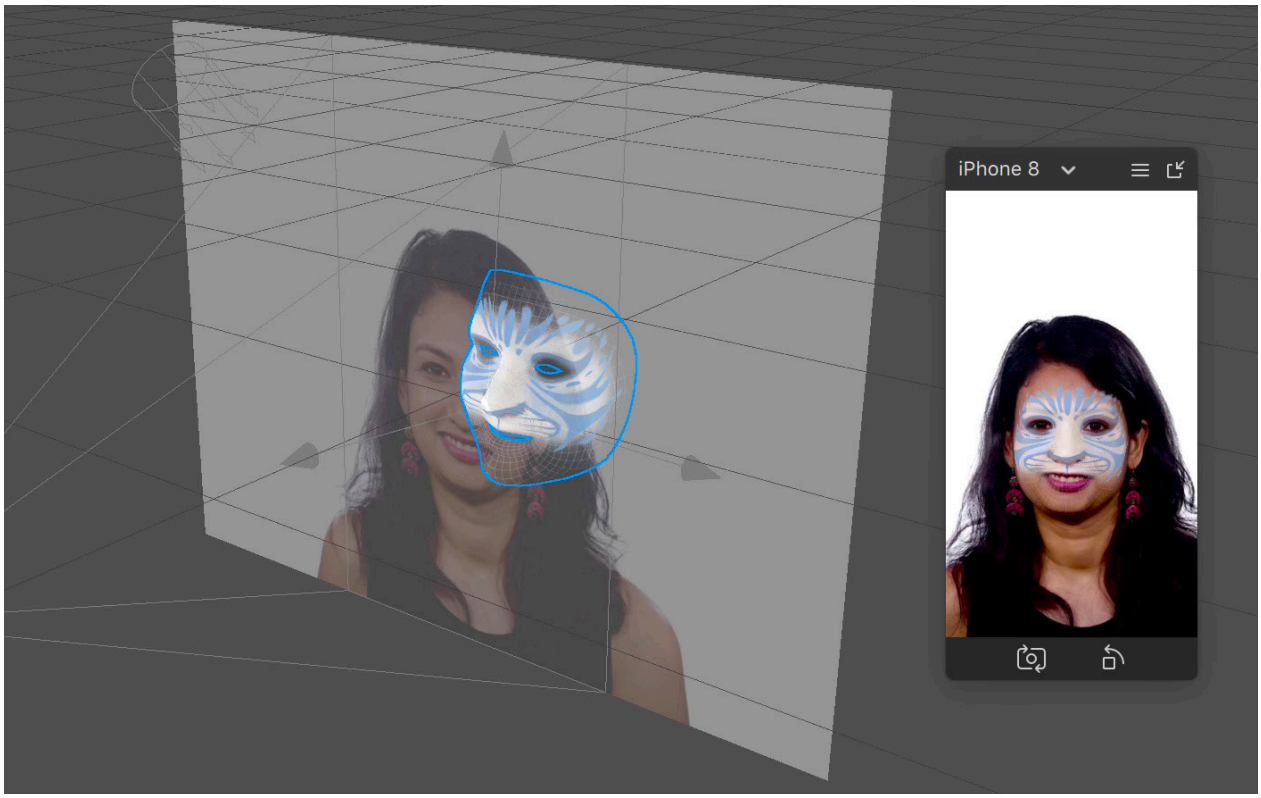
Face Landmark patches are used to capture the precise position of parts of the face, such as eyes and nose. This allows you to position objects like piercings to move in sync with facial and head movements.

Apart from objects positioned using the Eyeball patch, objects positioned with Face Landmark patches don't rotate in sync with the user's facial and head movements, but still move along the X, Y and Z axes as normal.

To position objects using Face Landmark patches, the object must be a child of a face tracker in the Scene panel.

These are the available Face Landmark patches:

Patch	Description
Cheek	Position an object on a precise location on the user's cheeks, such as the center of a cheek or a cheekbone.
Chin	Position an object on the tip of the user's chin.
Eyeball	Position an object on a location on the user's eyes (such as the center of an eyeball). This patch is used for iris tracking.
Eyebrow	Position an object on a precise location on the user's eyebrows, such as the middle of an eyebrow.
Eyelid	Position an object on a precise location on the user's eyelid (such as the inside or outside corner of an eye) and track the degree the eye is open.
Forehead	Position an object on the center or top of the user's forehead.
Mouth	Used to detect a specific part of someone's mouth.
Nose	Position an object on a precise location on the user's nose, such as a nostril or the tip of the nose.



Face tracker and face mesh

Using the face tracker

When a face tracker is combined with a face mesh, it creates a surface that can detect facial movements and expressions. You can add a material to the face mesh to create a mask effect.

Objects that are children of a face tracker will move with the face detected in the scene. To make an object the child of the face tracker, simply drag the object onto the face tracker in the Scene panel. You can then use the manipulators to adjust its position.

To track objects to precise positions on the face, use face landmark patches.

Pro Tip: Adding multiple face trackers to a scene will have an impact on performance, so it's always a good idea to test your effect as you build it using the Spark AR Player app.

HOW TO DYNAMICALLY MODIFY A SCENE

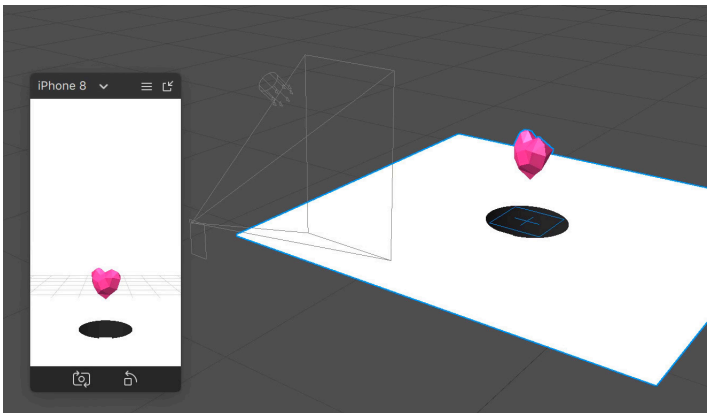
Using target markers

If your effect includes target tracking, it's a good idea to show a target marker in the effect until the target image is detected. This will help people using the effect know what target to look for in the real world. When the correct target is detected the target marker can be set to disappear, revealing the effect.

The target marker is made of 2 textures: a semi-transparent preview of the target image and a texture provided by Spark AR Studio that looks like a frame. The file is called **target-marker-texture.png** and you can download the image file here.

A simple combination of patches in the Patch Editor can make the target marker disappear when the target is found.

Each of these textures will be added to the scene using an object included in Spark AR Studio called a rectangle. The rectangle will automatically be added as a child of a canvas which will ensure it resizes in sync



Plane tracker

- The target image preview should be the same image that triggers the target tracking effect.
- To complete the effect, use the Patch Editor to make the target marker disappear when the target is found.
- Use Spark AR Player to preview your target tracking effect on a device

with the device screen. The canvas will also be used to group the target marker texture and the preview image, so changes can be made to both of them at the same time.

Using the plane tracker to integrate effects into the real world

A plane tracker finds a horizontal surface in the real world, like a floor to table. This allows you to place 3D objects in people's environments.

Plane tracking effects are only visible through the camera on the back of the device.

Once you select the plane tracker in the Scene panel, you're able to see its properties in the Inspector. The following are property options:

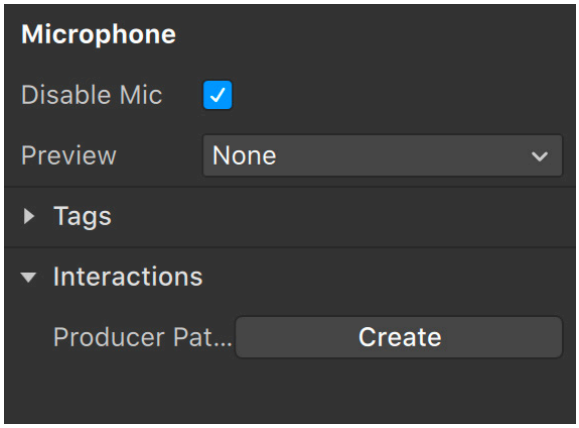
- **Layer** – click the dropdown next to Layer to adjust which layer the plane tracker is assigned to.
- **Visible** – clear this box to stop the plane tracker and any children from being rendered in the scene.
- **Auto Start** – select this box to find a plane in world space as soon as your effect is opened.
- **Transformations** – you can't change the transformations for a plane tracker, because it isn't an object.
- **Enable For** – this sets camera-specific visibility. For the plane tracker, Enable For is set to Back Camera automatically and can't be changed. This means that the plane tracker will only be visible in the back camera of a mobile device.

When adding a plane tracker to an effect, it's a good idea to add instructions so that users know that the effect should be viewed through the back camera. You can use the Patch Editor to do this.

The microphone

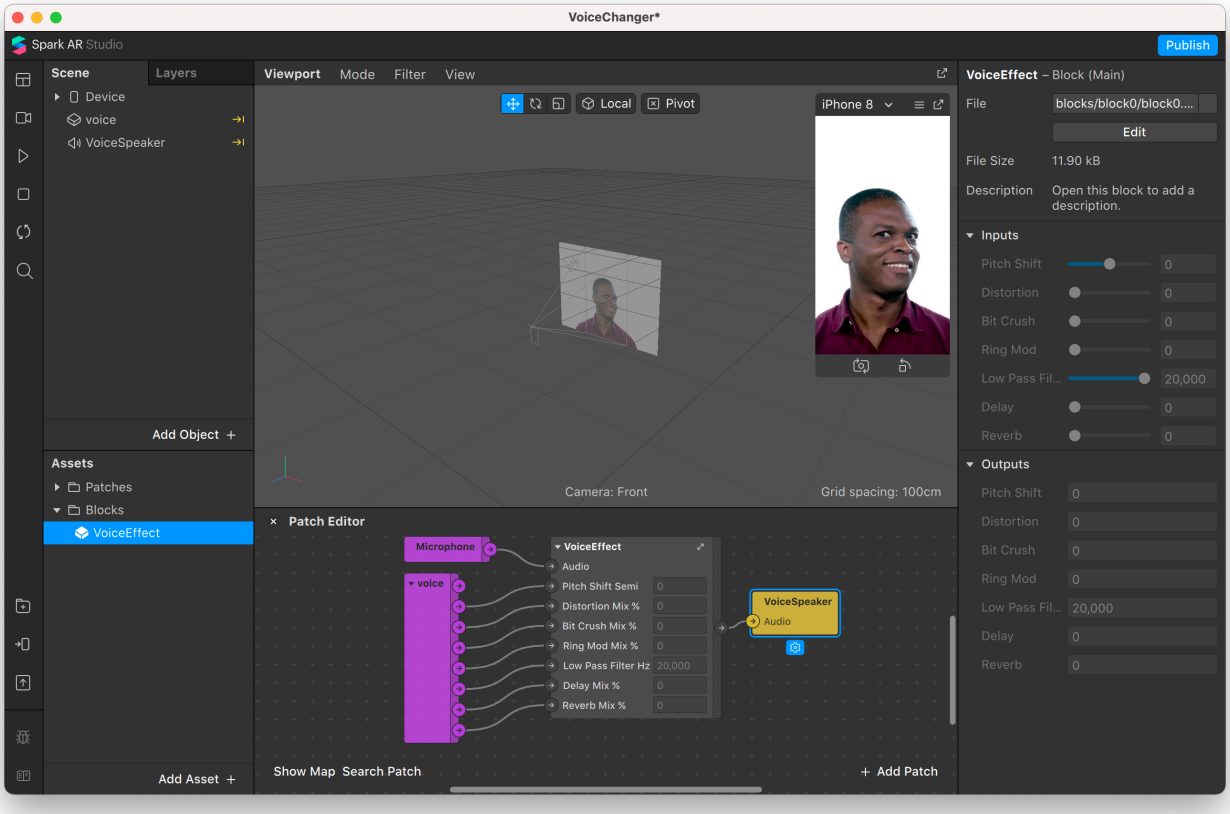
The microphone captures sound detected by the device's microphone when your effect is being used, and plays it back in the scene. You can use the microphone to analyze and modify the voice of the person using your effect.

By default the microphone records and plays back any sound detected when the effect is being used. Check the box 'disable mic' to stop microphone sound being recorded when capturing a video with the effect. Keep in mind that there are some limitations, for example, you can't play audio while at the same time recording with the microphone in an effect.



Microphone properties in the inspector panel

You can use audio effect patches to distort and modify the sound detected by the microphone in the Patch Editor. You can also analyze the sound detected by the microphone via the energy meter patch



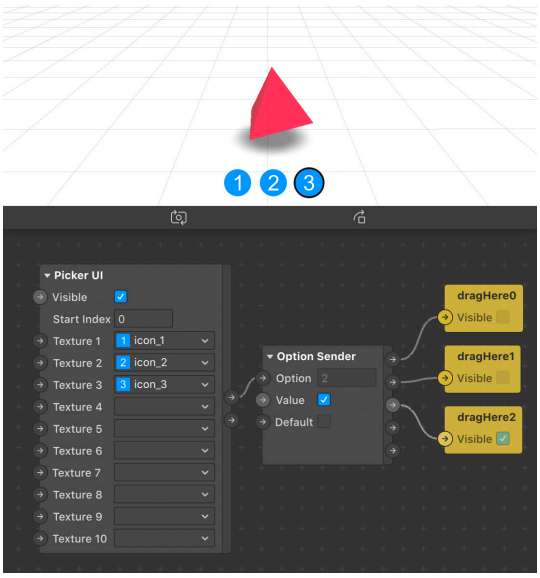
Voice changer template

How to use gallery textures

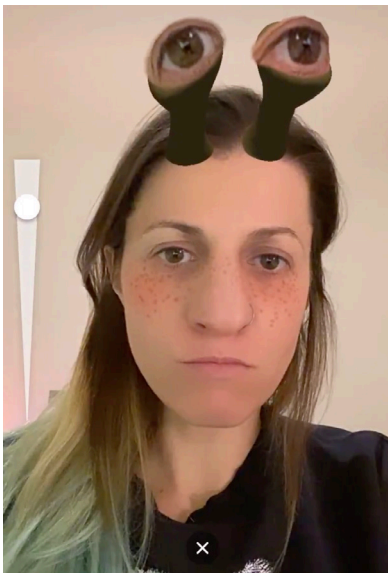
The Gallery Picker lets people add an image from their phone's camera roll to an effect, like a green screen. To create an effect with the gallery picker, all you need to do is add the gallery texture to a material in your scene. The material can be applied to any object, for example a 3D object or a rectangle to create a segmentation effect.

By default, the gallery texture renders as a blank white color. You can add a texture or color to the gallery texture, to be replaced with the image selected by the person using your effect.

If an effect includes the gallery texture, an 'add media' button will appear at the bottom of the effect. People can then browse and add images from their own media library.



Example of Picker UI



Example of Slider UI

There are a few ways you can edit the gallery texture's properties in order to change how it appears in the effect:

- **Pending texture** - checking this box lets you select an initial texture to show in the effect. This texture will be replaced with the image selected by the person using your effect.
- **Type** - a color or texture:
  - If the Type is set to Texture, add an image file here.
  - If the Type is set to color, select a color from the menu.
  - Used by - Any materials the texture is applied to will be listed here.

When planning effects that use the gallery texture, it's worth considering that only one camera roll texture can be selected by users at a time.

Either photos or videos can be added to effects with the gallery texture, and since those images or videos can be square, horizontal or vertical, you might need to resize the image to fit the area where you want to use it.

Processing user inputs

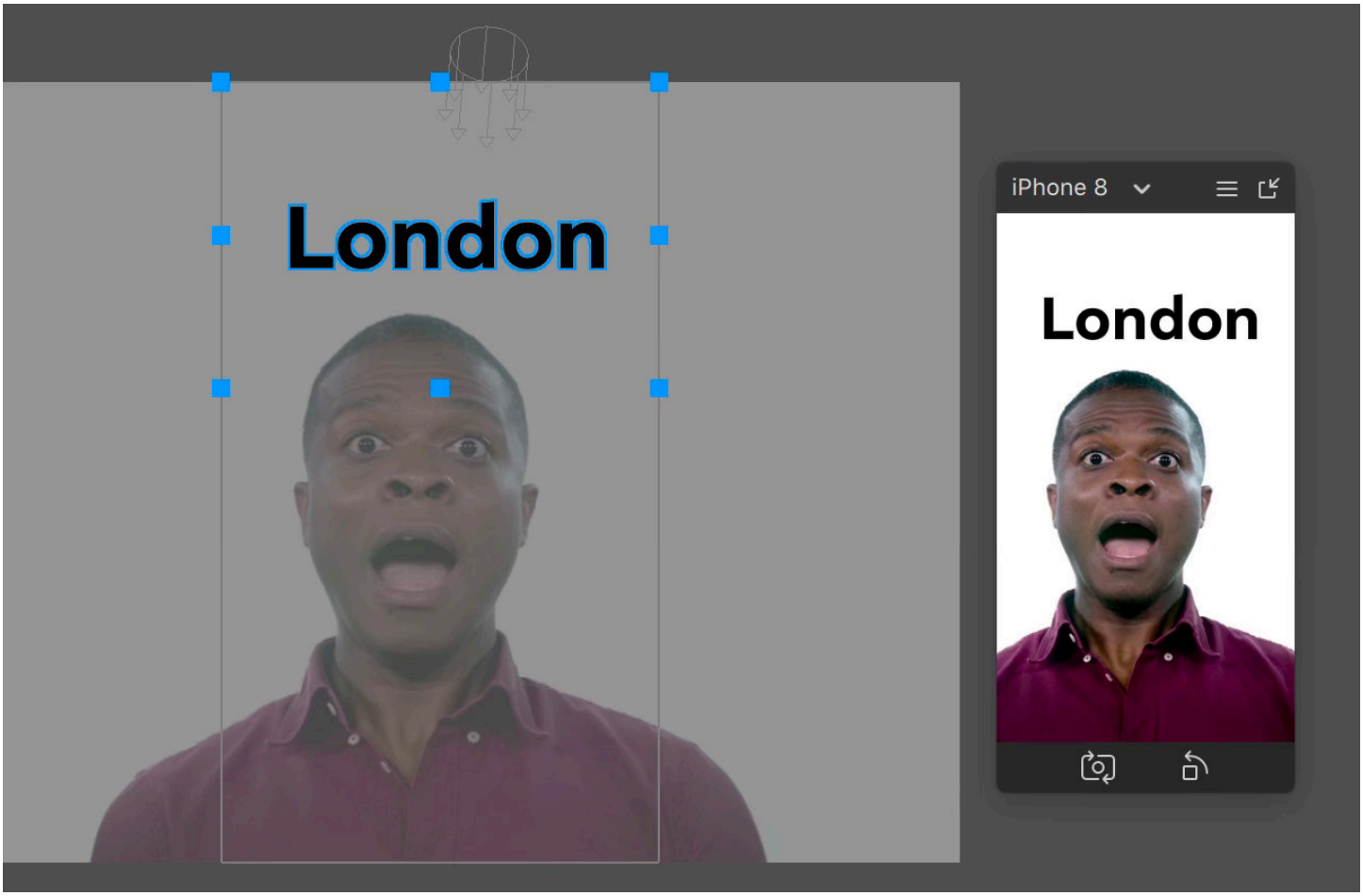
User interface patches let you add UI functions to your effect. These patches enable users to pick between different options or adjust properties in an effect by tapping on-screen icons or using an on-screen slider.

What is Native UI?

The native UI picker and slider are built-in interfaces that can be added to Spark AR effects. The UI picker - lets people choose different options by tapping icons on the screen. The UI slider allows users to increase or decrease some aspects of the effect.

The picker object needs to be configured with a list of uncompressed textures that are displayed in the picker.

Name	Description
Picker UI	Use the picker UI patch to select from up to 10 options that change something in your effect.
Slider UI	Enable people to change the value of something in your effect using an interactive slider.



Dynamic text

What is dynamic text?

Dynamic text is used to create personalized effects that show text specific to someone's situation. For example, their location or the date. You can add both 2D and 3D dynamic text to your effects.

TO MAXIMIZE THE USER'S EXPERIENCE, YOUR EFFECT SHOULD COME WITH INSTRUCTIONS.

Providing clear instructions helps people understand how to use your effect. Your instructions will appear in the middle of your device screen and you can set the duration time during your initial set-up.

You can add custom instructions by using **the Inspector** to show instructions when an effect has been open for a set amount of time. Or for more complex instructions involving logic and other interactions, you can use the **Patch Editor** or **scripting** to relay them to the user.



Instruction

Chapter 4

SCENE OPTIMIZATION



While it's important that your effects are engaging and high quality, they should also perform well on a variety of devices. In this chapter you'll learn a number of ways to improve the performance of your effect.

TECHNICAL GUIDELINES AND PERFORMANCE CONSIDERATIONS

There are maximum limits for object count, texture resolution and triangle count for 3D objects. You should follow the following guidelines in order to strike a balance between quality and performance.

- We recommended adding no more than 50 objects to a scene.
- The maximum resolution for a texture is 1024 px x 1024 px. Any textures you add to a project that are larger than this will be automatically resized.
- When importing 3D objects, please note that:
  - The maximum number of vertices per object should be below 20,000.
  - The total triangle count per object in an effect should be below 50,000.
  - Keep the total triangle count for all objects in an effect below 150,000.
  - The height should be between 1 centimeter (min) and 5 meters (max).
- Before exporting a project, make sure you've removed any capabilities that aren't used in the effect.

METHODS FOR OPTIMIZING ASSETS

Special considerations when creating 3D objects

If you create 3D objects with Blender, use Spark AR Toolkit to optimize your objects before exporting them to Spark AR Studio.

To reduce the impact of 3D objects with animations on your effect, use the least number of bones possible and avoid using complex blend shapes wherever possible.

Tips for optimizing textures

- Make textures as small as you can wherever possible.
- The maximum resolution for a texture is 1024x1024. Any textures you add to a project that are larger than this will be automatically resized.
- Make sure textures are square and sized to the power of 2 (example: 2x2, 4x4, 16x16, 32x32). Some texture compression options, like PVR, require textures to be in this format.
- Avoid long 2D texture sequences. If possible, render your sequence every other frame and avoid making 24 or 30-FPS sequences.

There are also limits to consider with segmentation effects.

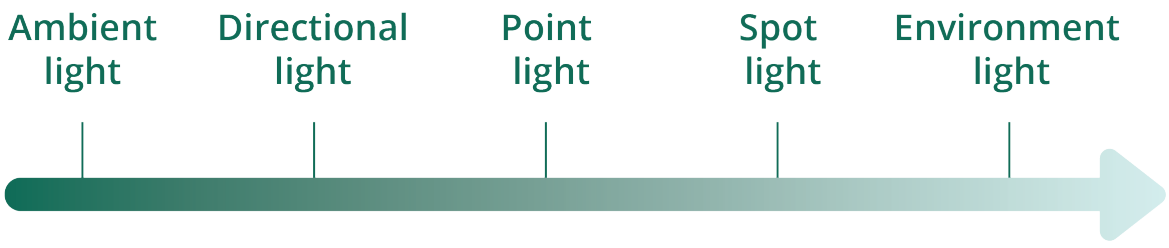
Segmentation in Spark AR Studio works best when detecting people from the chest upwards in the immediate foreground and in settings that have balanced lighting.

Segmentation only works on newer devices. iOS devices must be iPhone 6S or later and Android devices must be Samsung Galaxy S6, Sony Xperia Z2 or equivalent, or later.

Effects that contain both segmentation and the face tracker don't perform well. It's best to avoid using these capabilities within the same effect.

The more light sources in the scene, the higher the impact on performance.

Some light sources have more of an impact than others on the performance of an effect. Below, we've listed each light source in Spark AR Studio, starting with the lowest impact and ending with the highest:



It's a good idea to remove the ambient light automatically included in a project where possible. For example, if you're creating an effect with 2D objects that don't require realistic lighting.

METHODS FOR OPTIMIZING LOGIC

If you're using patches, grouping them can improve performance. You should also avoid duplicating logic — for example, calculating the same value more than once.

If you're using scripting, wherever possible avoid using subscriptions, cast variables as constant, and avoid scripts with lots of callbacks and functions for every frame.

How to use compression to build efficient effects

Before publishing an effect you should experiment with different compression settings to make your overall project size as small as possible. Smaller effects are quicker to load and this can improve the number of impressions your effect gets.

Any compression you apply in Spark AR Studio won't affect your original files.

When to use texture compression

To help decide which textures would benefit from compression, you can see the file sizes for textures in both downloaded and unpacked bundles for each device type (iOS, Android, older Android).

There are three possible compression settings in Spark AR Studio for each target device (iOS, Android, older Android):

- **Automatic** — This setting finds the best compression format for your texture on the targeted device, ensuring your effect works on all devices.
- **Manual** — Use this option to manually apply additional compression and control the settings.
- **None** — This setting isn't recommended but you could use it if you already compressed your textures before uploading them.

When to use image compression

When you choose the texture format, you're choosing to prioritize either file size or in-memory size. If you want a smaller effect file size, choose uncompressed. If you want a smaller in-memory size, choose ETC.

An uncompressed effect uses either a PNG or JPG image file. Using either one of these formats reduces the image file size when exporting the effect, but both of these files expand to the full raw image on loading.

ETC textures have a fixed compression ratio. This means that the image file can be larger than the equivalent PNG or JPG, but will be smaller than the full raw image. On loading, ETC textures remain the same size, which is smaller than the equivalent PNG or JPG.

TIPS FOR SAVING AND SHARING YOUR FILES

Use patch groups and patch assets

You can save and share groups of patches with other Spark AR Studio creators, and import ready-made patch groups into your own projects. It's a great way to efficiently build effects with logic and interactivity.

Any groups of patches you've added to a project will be saved in the patch library, under Saved Patches. You can share these files with other creators.

Once you've created a shareable patch asset file you can import or drag the .arp file directly into the Assets panel, like you would any other asset.

You can use the patch asset as many times as you want in your patch graph and quickly edit all instances of this patch asset at the same time.

Any changes you make to the patch asset, either in the Asset panel or in your graph, will be applied to the patch asset wherever it's used in the project.

> **Pro Tip:** A great way to find and share patch assets is by joining the Spark AR Creator Group. You can also import patch assets into your project directly from the AR Library to quickly add logic and interactivity to your effect.

Using project bundle to share your project

If you want to send a version of your project to somebody, you can bundle the project folder and file together to create a package.

Saving your project in a shared folder is the best way of collaborating over a period of time, but packaging projects is useful for quick and easy distribution of your work. To package your project:

- Click File in the menu bar.
- Select Package

Use AR blocks

With blocks, you can save and export sections of a project in Spark AR Studio, to be reused in other projects and shared with other creators.

You can create a block for most things you often need to repeat or reuse. Combinations of objects, assets, patches, materials and texture settings are all supported.

When you save a block, it'll automatically update in all projects that use the block. The quickest way to add a block to your scene is to simply click and drag it from the Assets panel to the Scene panel.

Inputs and outputs pass information to and from blocks, into your projects. You use them to set default values for the block's properties, allowing you to customize the block.

The following data types are supported:

- Pulse
- Boolean
- Scalar
- Vector2, Vector3 and Vector4
- Color
- Number
- Text
- Texture

It's a good idea to add descriptions to your blocks. This helps creators you share the block with understand what it does, and help you keep track too.

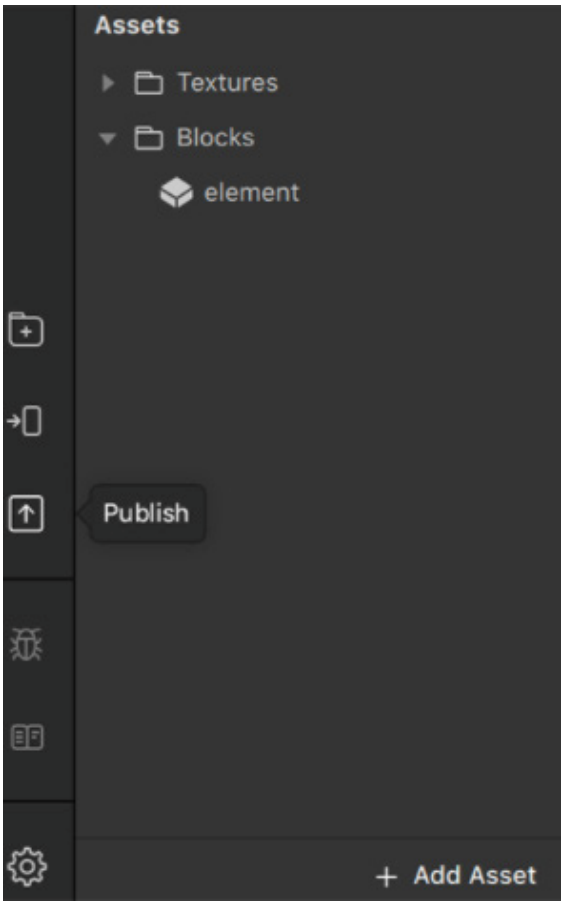
File size: what you should know

There are size limits for the files exported from Spark AR Studio to Spark AR Hub. These limits exist to ensure that your effects perform well across different devices. Said size limits are different depending on which platform you intend to make your effect available.

.arfx file size for effects according to platform

- **Instagram** – 4 MB or less on both iOS, Android and older Android.
- **Facebook** – under 2 MB for the best reach, but it can be up to 10 MB.
- **Facebook Ads** – under 5 MB

The .arexport file is the file that you upload to Spark AR Hub when you publish your effect and it contains the file that makes your effect available on each platform. It must always be 40 MB or smaller regardless of the platform.



Blocks in the asset panel

Make sure your effects are the correct size

You need to check your effect's file size across all three device types: iOS, Android and older Android. To do this, open your effect in Spark AR Studio and click the Publish icon in the toolbar.

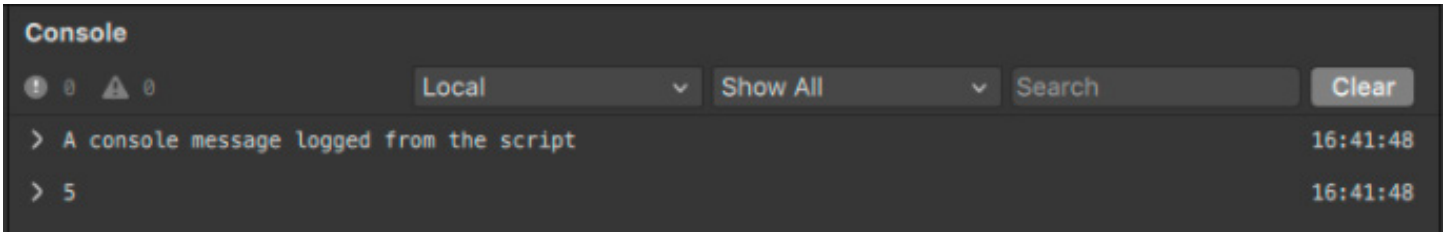
If you see the message 'File sizes meet requirements', your effect is the correct size for all device types and you can upload it to Spark AR Hub. If you see a cross next to the platform you want to make your effect available on, you need to make the file size smaller. There are lots of ways to reduce the file size, from removing unused assets to optimizing objects in your scene.

USING JAVASCRIPT TO TEST AND DEBUG SCENE INTERACTIONS AND PROCESS SIGNALS

The Console can be used to test and debug your projects in Spark AR Studio in 2 ways:

Console logging

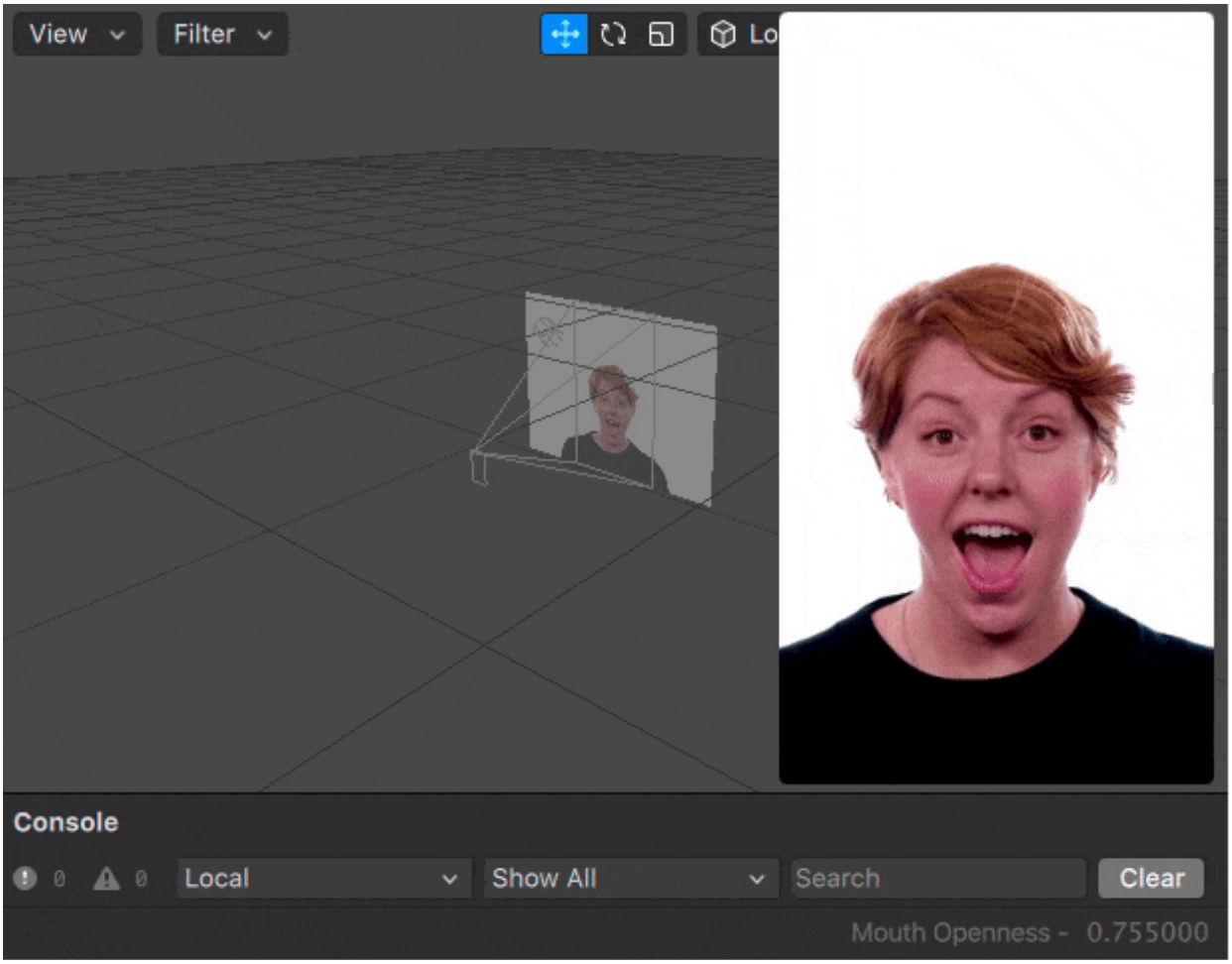
You can use the Console in Spark AR Studio to log messages and variables from scripts you've added to your project. Messages and variable values can be logged from within a script with the log() method of the Diagnostics Module.



Messages logged in the console

Signal watching

Signal values can be shown in the Console from within a script with the watch() method of the Diagnostics Module.



Using watch () method to monitor mouth openness value

Dynamically modifying a scene

With Spark AR Studio's dynamic instantiation feature you can create, destroy and reorder objects in the scene dynamically via script. Dynamic creation and destruction of scene objects, materials and blocks are all supported via calls to their respective APIs.

Dynamic objects aren't displayed in the UI by default. Instead, Spark AR Studio automatically detects when a dynamic API is called within a script and displays a notification prompting you to switch to dynamic mode.

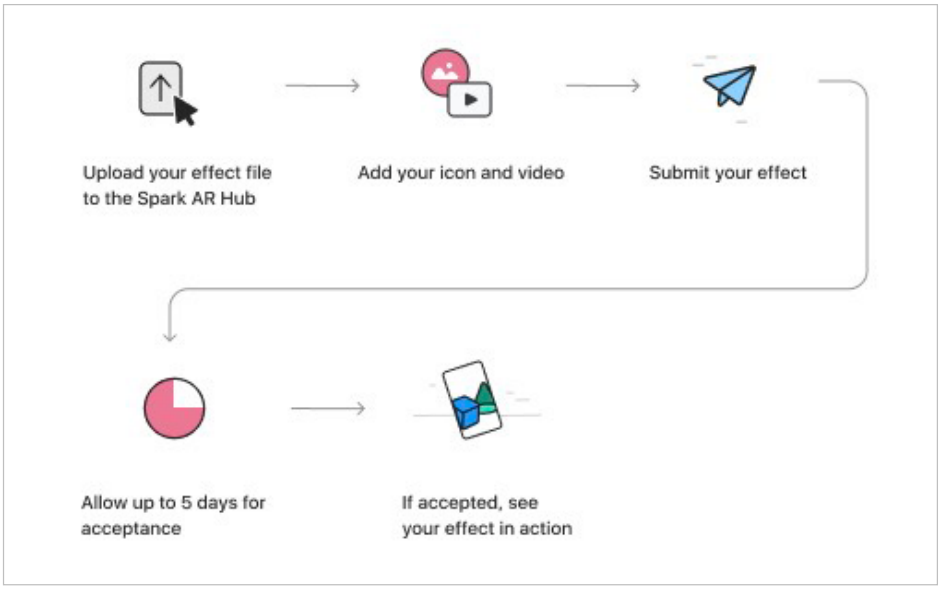
Note: Dynamic objects can't be edited from within the Spark AR Studio UI. Their properties are exclusively set via script.

Chapter 5

PUBLISHING AND EFFECT MANAGEMENT



Spark AR enables a global community of creators and developers to imagine, create and share augmented reality experiences that reach billions of people using the Facebook family of apps and devices. We want to create a place where you can make safe, respectful and high-quality effects and share them with the world.



HOW TO PUBLISH YOUR EFFECTS

During the publishing process, you'll need to select platforms, an owner and publishers. These options control where the effect is published, which account manages the effect and which account the effect is attributed to.

The publishing process for public effects includes:

- Choosing a name for your effect.
- Uploading your .arexport file.
- Choosing a platform, owner and publisher.
  - **Platform** – The platform is where you publish your effect.
  - **Owner** – The effect owner must either be a personal Facebook account or a Facebook Page. The owner manages the effect and controls permissions.
  - **Publisher** – The publisher controls the account your effect is attributed to. This means the profile or Page that will be shown next to the effect wherever it's displayed, used or shared. The publisher is either a Facebook account/ Page or an Instagram account linked to the effect owner.
- Selecting categories and keywords.
- Adding your demo video and icon.
- Choosing a publication date.
- Submitting your effect and waiting for it to be accepted.

Publishing upload workflow within the app

The publishing window from Studio offers the option to directly record a demo video that teaches people how to use your effect.

Once you choose whether you're publishing a new effect or updating an existing effect you can then upload directly to the Spark AR Hub. After the upload is complete, Spark AR Hub will open automatically in your browser.

OPTIMIZING AND EXPORTING EFFECTS

When you export an effect from Spark AR Studio, an .arexport file is saved to your computer. This is the file that you upload to Spark AR Hub when you publish your effect. It contains all the individual files needed for your effect to work on your chosen platform.

Exporting an AR export

The size limits for the files exported from Spark AR Studio to Spark AR Hub make sure your effects perform well across different devices.

- Instagram should be 4 MB or less on both iOS, Android and older Android.
- Facebook should be under 2 MB for the best reach, but they can be up to 10 MB.
- Facebook Ads should be under 5 MB.

PLATFORM REVIEW

To publish an effect you need to upload it to Spark AR Hub.

When working with clients you should let them know that it could take up to 5 business days for an effect to be reviewed and accepted.

If an effect is accepted, you'll be notified on Facebook and in Spark AR Hub. Depending on the destination you pick, your effect will then be publicly available through the app's camera.

Until the effect is accepted, it won't be visible to the public. However, you can share a link to the effect with your friends and followers.

Sometimes an effect is initially accepted but a further review finds that it doesn't meet the Spark AR policies. In this instance, your effect will remain visible but you'll need to update it within 10 days.

After the initial 10 days, if none of your updates have been successful or we're still reviewing your update, the effect will no longer be visible. The status of your effect will change to either:

- **Not Accepted** — you need to continue to submit updates to your effect and we'll notify you once it's accepted and visible again.
- **In review** — we're still reviewing your update and your effect won't be visible during this time.

If you don't want your effect to be public, you can also choose to publish your effect as a link only. Link only effects don't appear in the places that a public effect would, but are easier to publish and don't require a review.

Here's a scenario in which an effect you've created could be considered non-compliant:

You've created an effect for a client that promotes the use or depicts the sale of a potentially dangerous cosmetic procedure.

To encourage safety and well-being on our platforms, we don't accept effects that directly promote or encourage potentially dangerous cosmetic surgery. These include effects with instructional elements such as surgery lines or needles, as they clearly promote cosmetic procedures.

These depictions differ from effects that change the size and shape of facial features, known as face altering effects. For example, we allow effects that make the users' eyes or lips bigger or smaller. And also effects that are part of a character transformation — for example, those used to make the user look like an animal or fantastical character.

How to appeal a review decision

If your effect isn't approved, you'll be notified as to which issues need to be resolved. You can review the specific policies covering effect requirements in Spark AR Policies and either Community Standards or Community Guidelines.

To appeal a rejection, the rejected effect needs to be edited in Spark AR Studio and resubmitted for review.



Once uploaded to the Hub you can select whether to make your effect available on all supported platforms in the Facebook family of apps or on specific platforms only.

MAINTAINING, UPDATING AND IMPROVING EFFECTS

You can update your effects whenever you like, but sometimes we'll ask you to update older effects to make sure they keep working.

Effects will be supported for six months after the release of the version of Spark AR Studio you used to create them. After six months, effects might no longer be compatible with the latest versions of Instagram and Facebook, and people might not be able to find or use them.

If an effect needs to be updated you'll get an email or notification on Facebook. You'll also see a warning in Spark AR Hub.

You can make changes or updates to your effect after uploading it to the Spark AR Hub as well as updating the effect .arexport file. This applies for both published and unpublished effects.

SPARK AR AND FACEBOOK POLICIES

The Spark AR Policies are here to help you understand what type of content is and isn't allowed on the Spark platform. We encourage you to become familiar with Spark AR's Review Policies before submitting your content for publishing. The following are a few key highlights from our policy guides:

All content published to Facebook, Messenger or Portal must comply with our Community Standards, and all applicable laws, statutes and regulations.

All content published to Instagram must comply with the Instagram Community Guidelines, and all applicable laws, statutes and regulations.

All content created in Spark AR must be suitable for consumption by the Facebook, Inc. general user base (including those aged 13 and above). This means that content must **not**:

- Be shocking, sensational, disrespectful or excessively violent.
- Promote illegal products, services or activities, or content that is inappropriate or unsafe.
- Promote alcohol, tobacco, marijuana or pharmaceutical medical products, nor be published by brands associated with such products.

- Be of an adult or sexual nature — for example, nudity, depictions of people in explicit or suggestive positions, or activities that are sexually suggestive or provocative.
- Discriminate against, harass, provoke or disparage people.
- Promote the use or depict the sale of a potentially dangerous cosmetic procedure, as per the Facebook Community Standards. This includes effects that depict such procedures through surgery lines.

Chapter 6

PRACTICE: WORKING WITH CLIENTS

UNDERSTANDING THE NEEDS OF YOUR CLIENT AND TARGET AUDIENCE

Here is an example of what you can expect when working with a client brief.



Client brief

Client: Lucky Shrub

Goals: Showcase the benefits of a new customer-loyalty program and raise awareness.

AR Platform: Instagram

AR Experience: Face Effect

Target Audience: Age 16 – 40. Customers who are interested in home goods.

Tone: This should feel fun, exciting and new.

Additional Notes: The client wants the customer to think, feel and do. The customer should think about the client's products. They should feel entertained while interacting with an enchanted environment. The customer should want to use the effect more than once and share it with friends.

Potential Ideas:

Step 1: User opens the AR filter to find themselves in a foggy enchanted forest.

Step 2: The AR filter starts when the user opens their mouth. You can see magical spell particles fly out of the mouth. At the same time the fog clears to reveal the client's product in the background.

The Lucky Shrub Brand



BLANCH CAPS

Headlines
Uppercase
Tracking: 70% of headline size

ABCDEFGHIJKLMN
OPQRSTUVWXYZ
0123456789

How to analyze a brief and develop an effect based on client expectations.

A client brief contains a lot of information, but you need to know how to break it down. In this brief, client Lucky Shrub — an urban garden design shop and nursery that focuses on uniquely crafted green spaces and accessories for home and work— is looking for a creative way to showcase the benefits of their all-new customer loyalty program, and ultimately drive sign-ups. So the goal for this campaign is to create **awareness**.

Before getting started you'll want to get an idea for the brand's identity. Become familiar with their logos, typography and brand colors and consider how those can be integrated into your effect.

The brief has also detailed that the client wants an interactive effect that will show the benefits of the loyalty program. The client has expressed interest in an effect with magical and fantastical elements. They've also detailed that this effect will target their customer base on Instagram, which means that you'll need to keep in mind the limitations of Instagram Camera before starting your creative process.

Pro Tip: Clients care about making a good impression, so when designing your effect you should take into consideration selfie design and user interaction with the front-facing camera.

How to develop a creative strategy based on a client brief.

Evaluating a brief and developing a creative strategy before starting your design can be critical to how effective you will be at meeting the client's expectations. Think of your creative strategy as a guide that can help you focus on your goal and by which you can measure success.

In essence your strategy should focus on solving a problem for your client, and the effect that you will create should offer a solution. Some common problems brands might face can include:

- A brand feels like they aren't fun enough
- A brand wants to appear innovative
- A brand wants to encourage people to take action
- A brand wants to grow their audience on instagram

In order to create a creative strategy that is in line with your client's goals, it helps to ask yourself a few questions:

- What are the basic needs of the effect?
- What is the desired outcome for the brand?
- What is the desired outcome for the customer?
- Who stands to benefit from this and why?
- Is there an existing creative strategy and does this align?

GLOSSARY

Acronym	Term	Definition
	.arexport	Final zipped file that is created when you export an effect in Studio. 40 MB limit.
	.arproj	Project file (zipped JSON + internal assets). Instagram should be 4 MB or less on both iOS, Android and older Android. Facebook should be under 2 MB for the best reach, but they can be up to 10 MB. Facebook Ads should be under 5 MB
	.arp	File extension used when you make a patch asset.
	Ambient Light	Will provide each object in the scene with a constant amount of light. You can't change the position, rotation or scale of an ambient light.
	Animation Patch	Used to drive an animation triggered by an Interaction patch or logic.
	AR Effect	A piece of AR content created by a contributor/creator through Spark AR Studio, uploaded in Spark AR Hub, and used by users in Facebook, Instagram and Portal. This content allows users to create new content such as stories, videos, gifs, etc.
	Audio Clips	Audio clips should be mono M4A, AAC codec, with a sampling frequency of 44.1KHz. You can import your own or choose from a huge range of free, licensed sound effects in the AR Library.

Acronym	Term	Definition
	Audio Playback Controller	The audio playback controller renders sound in the scene only after it's connected to a speaker object. Use it to play sound continuously on a loop in your AR effect, or connect one-shot audio playback to boolean signals in the Patch Editor.
AR	**Augmented Reality**	An interactive version of reality created by the use of technology to overlay digital information on something being viewed through a device (such as a smartphone camera)
	Authoring	Authoring tools help you create digital content. They can be classified as programming or content design tools
	Blend Mode	Sets the blend mode of blend material: • Add • Associative Alpha • Alpha • Subtract • Multiply • Replace • Screen
	Blended Material	The blended material includes a blending mode, to mix textures and colors together.
	BG Influence	Multiplies a color or texture against a grayscale version of the video feed. Only available in the face paint material.
	Body Tracking	The body Tracking capability detects and tracks bodies in the scene.

Acronym	Term	Definition
	Brightness	Sets the brightness of a color. Only available in the face paint material.
	Camera	An object that's automatically listed in the Scene panel in every effect. It represents the camera in the device showing your effect, which means it can't be removed from a project.
	Camera Texture	A texture that extracts live video as it's being captured by the device camera. The camera texture can be combined with segmentation to separate someone from their background. You can also manipulate the pixels in the texture to modify the image.
	Canvas	A container for laying out 2D objects and making them responsive to the size of the device screen.
	Cheek Patch	Used to position an object on a precise location on the user's cheeks, such as the center of a cheek or a cheekbone.
	Chin Patch	Used to position an object on the tip of the user's chin.
Color LUTs	**Color Lookup Textures**	Color LUTs (Lookup Textures) are tables of RGB color values. In Spark AR, you can use color LUTs to quickly create color-graded effects across the entire scene.
	Cull Mode	Describes how material is culled: • **Front** – discard pieces of the mesh that face the viewer. • **Back** – Discard pieces of the mesh that face away from the viewer. Most effects use this mode to improve performance.

Acronym	Term	Definition
	Device	An object that's automatically listed in the Scene panel in every Spark AR Studio project. It represents the device showing the effect, which means it can't be removed from a project.
	Demo Video	Shows a preview of an effect to someone, giving them a flavor of what it will look like before they open it in the camera. On Instagram, they appear alongside your effect in the effect gallery and in the effect tray.
	Directional Light	Will shine on each object in the scene from the same direction, like the sun. It's a good idea to have a directional light in any scene that contains 3D objects, as they give objects a sense of depth.
	Double-sided	Displays both the front facing and back-facing sides of the mesh
	Dynamic Text	Use dynamic text to create personalized effects that show text specific to someone's situation. For example, their location or the date.
	Effect Revision	Each effect has a number of revisions, think of this as the versions of the effect. For example a contributor/creator uploads the effect for the first time, this is the first revision. Then the name/icon/demo-video changes, this will create a second revision for that effect with all the updated information.
	Effect Owner	The personal Facebook account or Facebook Page that manages the effect and controls permissions.

Acronym	Term	Definition
Color LUTs	**Enumerated**	A data type that consists of predefined values. Enums provide a highly-structured way to store data since they can only store a single predefined value.
	Environment Texture	Used to mimic the light in a specific real world environment. This is also known as using image-based lighting.
	Eyeball Patch	Used to position an object on a precise location on the user's eyes. This patch is used for iris tracking.
	Eyebrow Patch	Used to position an object on a precise location on the user's eyebrows, such as the middle of an eyebrow.
	Eyelid Patch	Used to position an object on a precise location on the user's eyelid, such as the inside or outside corner of an eye. This patch also allows you to track the degree the eye is open.
	Face Landmark Patch	Used to capture the position of parts of the face, such as eyes and nose. This allows you to position objects like piercings to move in sync with facial and head movements.
	Face Mesh	The face mesh is a 3D model of a generic face. It works in combination with the face tracker in Spark AR Studio to create a surface that mimics someone's expressions.
	Face Paint Material	Use the face paint material to create a mask effect that shows the skin and features of the person using the effect - behind either a custom texture or a color.

Acronym	Term	Definition
	Face Reference Assets	The Face Reference Assets are a collection of textures and 3D objects. Use them when you're creating face effects in Spark AR Studio.
	Face Tracker	The Facetracker is a smart object that detects and responds to the position and orientation of a face.
	Face Tracker Texture	Allows you to use the video of someone's face when they're using the effect as a texture in your scene. You can apply the texture to a material and object – for example, to create a face swap effect.
	Feature tracking	Tracks the perspective (position, rotation, viewing angle, scale etc) of a predefined 2D pattern - typically an image. Works by tracking the 2D translation of individual feature points and solving for the change in perspective from frame to frame.
	Flat Material	This material is more performant than the standard material and doesn't respond to lighting.
	Forehead Patch	Allows you to position an object on the center or top of the user's forehead.
	Gallery Picker	Lets people add an image from their phone's camera roll to an effect, like a green screen.
	Hand Tracker	A tool that finds and follows a single hand in your scene and controls the position of any child objects.

Acronym	Term	Definition
HSL		The HSL (for hue, saturation, lightness) represents how colors appear under light.
HSV		HSV stands for hue, saturation, value. It's also commonly referred to as HSB, or hue, saturation, brightness. These are alternative representations of the RGB color model. In these models, colors of each hue are arranged in a radial slice around a central axis of neutral colors which ranges from black at the bottom to white at the top.
	Icon	An image that represents your effect in the camera, and can be either an image you upload or a frame from your demo video. You can upload this image to the Spark AR Hub while publishing your effect.
	Inspector Panel	Inspector panel is part of the Spark AR Studio interface. It's the panel at the right-hand side where we can inspect and change the properties of objects.
	Linear Space	Linear Space is the color and luminance values distributed in a straight line from bottom left to upper right
	Microphone	Captures sound detected by the device's microphone when your effect is in use and plays it back in the scene. You can use the microphone to analyze and modify the voice of the person using your effect.
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	Mouth Patch	Allows you to detect a specific part of someone's mouth.
	Nose Patch	Allows you to position an object on a precise location on the user's nose, such as a nostril or the tip of the nose.
	Null Object	A container that is used to group objects. It doesn't have a shape, but it can be manipulated in 3D space.
	Occluders	Occluders make AR effects more realistic by hiding things that wouldn't be visible in real life. Occluders are made by using an object and a material.
	Opacity	Sets the transparency of the material.
	Patches	Allow you to add interaction, animation and logic to effects. Patches act as visual building blocks, with each patch performing a different function. Connected together in what are known as a graph, patches pass and receive information to and from one another to make things happen in the effect.
	Patch Asset Material	Visual shaders created in the Patch Editor then set as a Shader Type for a material in the Inspector.
	Patch Editor	Implements visual programming. • Lowers the barrier to entry to creators without a technical background • Increases speed of creation of interactive effects • Supports bridging with JS scripting

Acronym	Term	Definition
	Particle Systems	Let you display and move large numbers of objects, called particles. You can apply force and drag particles to mimic the effects of gravity. Use particle systems to produce all kinds of visual experiences. For example, rain, smoke and confetti.
	Plane	A flat 3D object that can be placed at any depth within the scene. You could use a plane to place a 2D image or texture into 3D space
	Plane tracker	A tool that tracks an infinite, horizontal plane. Currently, Spark AR studio supports only one plane tracker in the scene at a time.
	Platform	The platform is where you publish your effect.
	Point Light	A sphere-shaped light that shines in all directions evenly. It becomes less intense with distance, like a lamp.
	Point tracker	Tracks a specified point.
	Physically-based Material	These materials are used to create realistic-looking objects in Spark AR Studio. The different properties in this material allow you to add surface roughness, metallic effects and lighting that mimics real world light.
	Publisher	The publisher is either a Facebook account/ Page or an Instagram account linked to the effect owner and controls the account your effect is attributed to.

Acronym	Term	Definition
	Reactive Programming	Spark AR Studio uses a reactive programming model, a declarative paradigm which is primarily structured around asynchronous data streams and the propagation of change. The reactive programming model allows you to define relationships between objects and values in such a way that a value bound to a signal is automatically updated when the latter changes.
	Real-World Scale	Allows your object to be sized to fit in the real world. For example, if your effect allows users to place items from a furniture catalog in their living space, the 3D furniture should be sized based on the real-world space. There are 2 ways to enable real world scale in Spark AR Studio depending on the type of user experience you want to create: • Enable Real-World Scale in the Inspector. Your object appears realistically scaled but only after a real-scale signal is detected. This means the user sees the object abruptly change size at some point after opening the effect. • Enable Real-World Scale in the • Inspector and add further logic via the Patch Editor or scripting. This method allows you to hide the object until a real-scale signal is detected. This means the user only sees the object once it's realistically scaled.
	Rectangle	A rectangle is a 2D shape in Spark AR Studio. Use it to render textures and materials, creating 2D effects like frames, backgrounds and colored overlays.

Acronym	Term	Definition
RGB		RGB means Red Green Blue channel. An RGB file consists of composite layers of Red, Green and Blue, each being coded on 256 levels from 0 to 255.
	Retouching Material	Used to add retouching effects to faces and scenes.
SVG	**Scalable Vector Graphic**	A graphic file type that is particularly useful in AR effects, because they don't lose precision when they're rescaled.
	Script/Scripting	A program or sequence of instructions that are interpreted or carried out by another program.
	Scripting API	Serves as a basis for using coding language, in this case Javascript, to execute programs.
	Segmentation	Used to identify and separate parts of the camera input. For example, to separate someone's hair from the rest of the scene so you can change the color without changing anything else in the effect.
	Shader Asset	A material type resulted from shader code.
	Shader Code Asset	The shader code asset lets you write custom shaders in Spark AR Studio.
	Signals	Signals are special objects that contain a value that changes over time.
	Simple Shadows	Adds a shadow to a 3D object to make it look more realistic.
	Simulator	The simulator represents a screen. Use it to preview how your effect will look on a device.

Acronym	Term	Definition
Spark SL	**Spark Shader Language**	SparkSL is Spark AR's own shading language that allows you to program your own shaders.
	Speaker	The speaker is an object in Spark AR Studio that renders sound in your effect. You can add multiple speakers to a project to render different audio clips.
	Spot Light	A cone-shaped beam of light, like a flashlight or headlight.
	sRBG Space	sRGB is the standard RGB (red, green, blue) color space and one of the options color-encoded options supported in Spark AR
	Standard Material	This material uses a lighting system called the Phong model that's good for simulating realistic lighting on 3D objects.
	Target Marker	Helps people using the effect know what target to look for in the real world.
	Target Tracker	Spark AR Studio allows creators to create effects that get triggered when the camera points to an image in the real world, for example a movie poster. This "target" image is then tracked in the world using SLAM (simultaneous localization and mapping).
	Touch Gestures	Are all the touch gestures made by the user, it includes, screen tap, object tap, long press, pan, pinch, and rotate.
3D	**Three-dimensional**	Describes an image that has the perception of depth. For example, a shape or object that has three dimensions: length, width and height.

Acronym	Term	Definition
	Three-dimensional (3D) Object	A 3D model or asset created in another program that can be imported into Spark AR.
	Tracking	Tracking answers the question “where is the object of interest located in the camera frame over time?”
2D	**Two-dimensional**	A flat-plane figure with only 2 dimensions, such as width and height, but lacking depth.
UX		Short for user experience.
UI		Short for user interface.
	UV Map	A UV map is a 2D representation of the surface of your 3D object.
	Vector Objects	Used to render a scalable vector graphic (SVG) image in your scene.
	Viewport	The viewport is where you can see and build your effect.
	Visual Programming	Allows users to create effects by connecting patch nodes graphically rather than scripting them in code.
	Visual Shaders	A shader is a set of algorithms that determines the appearance of a 3D object’s surface. They’re particularly useful for applying complex changes to a material.

Congratulations!

You've reached the end of the study aid.